

# CHAPTER 1

## INTRODUCTION

In the modern digital era, educational institutions are required to manage a large volume of academic and administrative data, including student admissions, faculty allocation, examination processes, and performance evaluation. With the increasing number of students and institutional activities, traditional manual systems are no longer efficient in handling these operations. Manual record-keeping, disconnected systems, and lack of automation often lead to data redundancy, processing delays, and increased chances of human errors. Additionally, the absence of proper integration between different departments makes it difficult to maintain consistency and transparency in academic management. Although some digital systems exist, they are often limited in functionality and do not provide a complete solution for managing institutional workflows.

To overcome these challenges, this project proposes a **Web-Based Smart ERP Solution for Academic Institutions**, which integrates all major academic and administrative functions into a single centralized platform. The system is designed to automate key processes such as student admission, faculty management, timetable generation, examination handling, and performance analysis. By incorporating advanced technologies such as automation, artificial intelligence, and data analytics, the system ensures efficient data processing, reduces manual workload, and improves overall productivity. The proposed solution not only simplifies institutional operations but also enhances communication between students, staff, and administrators. It provides a user-friendly interface and intelligent features that support better decision-making and efficient academic management.

## 1.1 PROBLEM DEFINITION

In today's educational institutions, managing academic and administrative activities is a complex and time-consuming process. Tasks such as student admission, document verification, cutoff calculation, seat allocation, faculty assignment, timetable scheduling, examination management, and performance tracking are often handled manually or through separate systems. This lack of integration leads to inefficiencies, duplication of data, and delays in processing. Manual handling of data increases the chances of human errors, affecting the accuracy and reliability of institutional records.

Moreover, communication between students, staff, and administrators is often not streamlined, leading to confusion and delays in information sharing. Traditional systems also lack advanced analytical capabilities to evaluate student performance, identify weak students, and provide insights for improvement. Evaluation of answer scripts is another challenging task, especially for descriptive answers, as it requires significant time and effort from faculty members. Existing systems do not effectively utilize modern technologies such as artificial intelligence and data analytics to improve these processes.

As a result, institutions face difficulties in maintaining data consistency, improving operational efficiency, and making informed decisions. Therefore, there is a need for an integrated and intelligent system that can automate academic processes, ensure secure data management, improve communication, and provide analytical insights to enhance overall institutional performance.

## 1.2 PROJECT DESCRIPTION

The proposed project aims to develop a **Web-Based Smart ERP Solution for Academic Institutions** that provides a unified platform for managing academic and administrative activities efficiently. The system integrates multiple modules, including admission management, administrative control, HOD management, connection platform,

AI-based evaluation, and analytical module, to ensure smooth and effective institutional operations.

The admission module allows students to register through the system by submitting their personal details and required documents. The system automatically verifies the data, calculates cutoff marks, and processes applications based on predefined criteria. It also enables automatic or manual assignment of guides and communicates updates to students and staff through notifications. Once the admission process is completed, the system allocates seats based on student preferences and merit, ensuring transparency and fairness.

The administrative module provides centralized control over all institutional data, including student records, staff information, and academic details. It ensures secure data handling and allows administrators to manage and update information efficiently. The HOD module enables department heads to assign faculty to subjects based on workload and generate timetables automatically using optimization techniques, ensuring conflict-free scheduling.

The system also includes a connection platform where students and staff can interact through quizzes, coding tests, and assessments. A reward-based system using coins is introduced to encourage student participation, where students can earn rewards and access academic benefits with necessary approvals. The AI-based evaluation module uses Optical Character Recognition (OCR) to extract text from answer scripts and applies keyword matching algorithms to evaluate responses automatically. Faculty members can review and modify the results if needed before final submission.

Additionally, the analytical module provides advanced visualization tools such as charts and dashboards to analyze student performance. It helps in identifying high-performing students as well as those who are at risk, enabling timely intervention. Students can also view their performance and compare it with others, promoting self-improvement. Overall, the system enhances efficiency, reduces manual effort, and improves academic management through automation and intelligent processing.

## 1.3 PROPOSED SYSTEM

The proposed system is a web-based application that can be accessed through any standard web browser on desktops, laptops, or mobile devices. It is designed to automate and manage academic and administrative activities in educational institutions by integrating multiple functionalities into a single platform. The system ensures efficient data handling, secure access, and smooth communication between users, including students, staff, HODs, and administrators.

The system consists of the following key components:

- **User Interface (Frontend):** This is the part of the system that users directly interact with. It provides a simple and user-friendly dashboard where users can perform various activities such as registration, viewing information, attending tests, and analyzing performance. The interface is designed to present data clearly and help users easily navigate through different features.
- **Backend Logic (Server-side):** The backend acts as the core processing unit of the system. It handles all operations such as data processing, authentication, and communication between different modules. It ensures that user requests are processed efficiently and the results are returned accurately.
- **Admission Module:** This module manages the entire admission process, including student registration, document submission, cutoff calculation, and seat allocation. It automates the process and ensures transparency and fairness.
- **Administrative Module:** This module provides centralized control over institutional data. It allows administrators to manage student records, staff information, and academic data securely.

- **HOD Module:** This module helps department heads assign faculty to subjects and generate timetables automatically based on constraints such as workload and availability.
- **Connection Platform:** This module enables interaction between students and staff through quizzes, coding tests, and academic activities. It also includes a reward-based system to encourage participation.
- **AI-Based Evaluation Module:** This module automates the evaluation of answer scripts using OCR and keyword matching techniques. It reduces manual effort and improves evaluation efficiency.
- **Analytical Module:** This module provides insights into student performance through charts and dashboards. It helps in identifying trends and making data-driven decisions.

### 1.3.1 Advantages of Proposed System

The proposed system offers several advantages that significantly improve the functioning of educational institutions:

- **Improved Efficiency:** Automation of processes such as admission, timetable generation, and evaluation reduces manual effort and speeds up operations.
- **Reduced Human Errors:** Automated processing minimizes errors in data handling and calculations.
- **Centralized Data Management:** All institutional data is stored and managed in a single system, ensuring consistency and easy access.
- **Enhanced Communication:** The system improves communication between students, staff, and administrators through notifications and updates.
- **AI-Based Evaluation:** Automatic evaluation of answer scripts saves time and ensures consistency in grading.
- **Performance Analysis:** Analytical tools help in monitoring student performance and identifying areas for improvement.

- **User-Friendly Interface:** The system provides an intuitive interface that is easy to use for all users.

### 1.3.2 Limitations of Proposed System

Despite its advantages, the proposed system has certain limitations:

- **Dependence on Data Accuracy:** The system relies on accurate input data for proper functioning. Incorrect data may affect results.
- **AI Evaluation Limitations:** The AI-based evaluation module may not always be fully accurate for complex descriptive answers and may require manual verification.
- **Technical Dependency:** The system requires proper internet connectivity and technical infrastructure for smooth operation.
- **Initial Setup and Training:** Users may require initial training to effectively use the system.

## CHAPTER 2

### LITERATURE SURVEY

#### **2.1 Title: A Comprehensive Study on ERP Systems in Higher Education**

**Authors:** A. Alshawhi, M. Themistocleous

**Year of publication:** 2020

#### **Description:**

This paper discusses the role of Enterprise Resource Planning (ERP) systems in educational institutions and how they help in integrating academic and administrative processes into a unified system. The study highlights how traditional institutions rely on fragmented systems for managing admissions, student records, faculty allocation, and examination processes, which leads to inefficiencies and data inconsistencies. ERP systems provide a centralized platform where all data is stored and accessed, improving coordination between departments. The paper also explains how ERP systems enhance transparency, reduce redundancy, and improve decision-making through better data availability. It emphasizes modular architecture where different components such as finance, academics, and administration are integrated into a single framework.

#### **Limitations Identified:**

- High implementation cost and complexity
- Lack of customization for specific institutional needs
- Limited analytical and predictive capabilities
- Difficulty in integrating advanced AI-based features

#### **How Our Project Addresses These Limitations:**

The proposed Smart ERP System is designed to be lightweight and modular, reducing implementation complexity. Unlike traditional ERP systems, it includes AI-based evaluation and analytical modules, which enhance decision-making and automation. The system is customizable and specifically designed for academic institutions, ensuring better adaptability and usability.

## **2.2 Title: Automated Student Evaluation System Using Machine Learning Techniques**

**Authors:** R. Kumar, S. Sharma

**Year of publication:** 2020

### **Description:**

This paper focuses on automating the student evaluation process using machine learning techniques. It explains how traditional evaluation methods are time-consuming and prone to inconsistencies. The proposed system uses pattern matching and classification algorithms to evaluate answers automatically. The study highlights the use of text similarity measures and keyword extraction techniques to compare student answers with reference answers. The system improves grading speed and consistency while reducing manual effort. It also discusses the potential of AI in transforming academic evaluation systems.

### **Limitations Identified:**

- Limited accuracy for descriptive answers
- Difficulty in understanding context and semantics
- Requires large datasets for training
- Lack of human validation mechanisms

### **How Our Project Addresses These Limitations:**

Our project combines AI-based evaluation with manual verification, allowing staff to review and modify marks if needed. This hybrid approach improves accuracy and reliability. The system uses keyword matching and structured evaluation, making it efficient even without large datasets. It ensures better balance between automation and human control.

## **2.3 Title: Automated Timetable Generation Using Genetic Algorithms**

**Authors:** S. K. Jain, P. Sharma

**Year of publication:** 2019

### **Description:**

This paper presents a method for automatic timetable generation using genetic algorithms. The system considers constraints such as faculty availability, classroom allocation, and subject requirements to generate conflict-free timetables. Genetic algorithms are used to optimize scheduling by iteratively improving solutions. The study shows that automated timetable generation reduces manual effort and ensures efficient resource utilization. It also improves scheduling flexibility and reduces human errors.

### **Limitations Identified:**

- High computational complexity
- Requires fine-tuning of parameters
- Difficult to implement in real-time systems
- Limited adaptability to dynamic changes

### **How Our Project Addresses These Limitations:**

Our system uses simplified rule-based and constraint-based algorithms for timetable generation, making it more practical and easier to implement. It supports dynamic updates and allows manual adjustments, ensuring flexibility and real-time usability.

## **2.4 Title: Student Performance Analysis Using Data Mining Techniques**

**Authors:** K. Ramesh, V. Priya

**Year of publication:** 2022

### **Description:**

This paper explores the use of data mining techniques to analyze student performance and predict academic outcomes. It discusses how educational institutions can use historical data such as marks, attendance, and test scores to identify trends and patterns. Techniques such as classification, clustering, and regression are used to analyze data and predict student performance. The system provides insights that help educators identify weak students and improve teaching strategies.

### **Limitations Identified:**

- Requires large and clean datasets
- Complex implementation of algorithms
- Limited real-time analysis capabilities
- Lack of user-friendly visualization

### **How Our Project Addresses These Limitations:**

Our system provides real-time analytical dashboards using simple visualization techniques such as charts and graphs. It does not rely on complex data mining algorithms, making it easier to implement. It allows both students and staff to easily understand performance insights without requiring technical expertise.

## **2.5 Title: Web-Based Online Examination System**

**Authors:** N. Gupta, A. Verma

**Year of publication:** 2020

### **Description:**

This paper describes a web-based online examination system that allows students to take tests and receive results digitally. The system supports multiple types of questions, including objective and subjective questions. It automates result calculation and provides instant feedback to students. The system improves efficiency and reduces the need for manual paper-based examinations. It also ensures transparency and accessibility in conducting exams.

### **Limitations Identified:**

- Limited support for coding-based evaluations
- Lack of integration with other academic modules
- No advanced reward or engagement system
- Minimal analytical features

### **How Our Project Addresses These Limitations:**

Our system extends beyond basic examination systems by integrating quizzes, coding tests, and a reward-based coin system. It also connects with other modules such as admission, evaluation, and analytics, providing a complete ERP solution. This improves engagement and overall system functionality.

## **2.6 Title: Optical Character Recognition Based Automated Grading System**

**Authors:** D. Singh, R. Mehta

**Year of publication:** 2020

### **Description:**

This paper presents an OCR-based system for automatically grading handwritten answer scripts. The system converts scanned documents into text using OCR and evaluates them using predefined answer keys. It reduces manual effort and speeds up the grading process. The study highlights the importance of OCR in educational automation and discusses challenges in handling handwriting variations.

### **Limitations Identified:**

- OCR accuracy issues with handwritten text
- Difficulty in handling complex answers
- High dependency on image quality
- Limited integration with other systems

### **How Our Project Addresses These Limitations:**

Our system integrates OCR with keyword-based evaluation and allows manual verification by staff, improving accuracy. It is also part of a larger ERP system, ensuring seamless integration with other modules such as analytics and student management. This makes the system more robust and practical.

## CHAPTER 3

### SYSTEM REQUIREMENTS SPECIFICATION

#### 3.1 HARDWARE REQUIREMENT

The hardware requirements for the Web-Based Smart ERP Solution are minimal, as the system is designed to be lightweight and accessible through a web browser. The application can be developed and deployed using standard computing devices without the need for high-end infrastructure. The system can efficiently run on the following hardware specifications:

- Laptop or Personal Computer
- Processor: Intel i3 or higher
- RAM: 4 GB or higher
- Storage: 100 GB or higher
- Internet Connectivity: Required for accessing the web application and database services

Since the system is web-based, users can access it from any device such as desktops, laptops, or mobile devices with an internet connection. A high-performance server is not mandatory for initial deployment; however, it can be considered in the future to support a larger number of users, faster processing, and better scalability. The hardware requirements ensure that the system remains cost-effective and easy to implement in educational institutions.

## 3.2 SOFTWARE REQUIREMENT

The software requirements for the Web-Based Smart ERP Solution include various tools and technologies used for developing, deploying, and managing the system efficiently. These technologies are selected to ensure reliability, scalability, and ease of use.

- Operating System: Windows 10 or higher
- Programming Language: Java / Python
- Backend Framework: Spring Boot (Java) / Flask (Python)
- Frontend Technologies: HTML, CSS, JavaScript
- Database: MySQL / MongoDB
- APIs: REST APIs for communication between frontend and backend
- Authentication: Role-based authentication system for secure login
- Development Tools: Visual Studio Code / IntelliJ IDEA
- Version Control: Git and GitHub

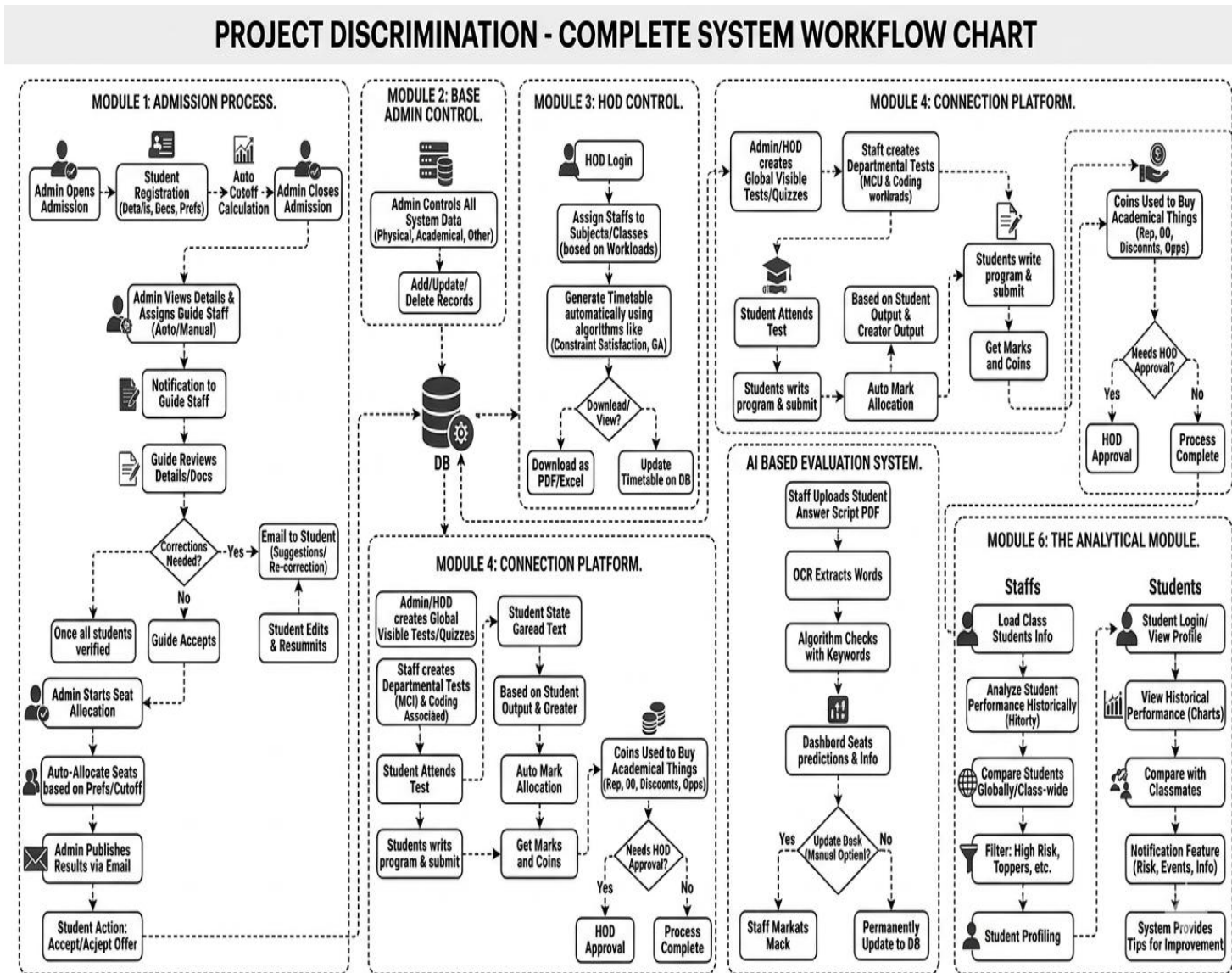
The backend technologies are used to handle business logic, data processing, and communication between different modules. The frontend technologies provide a user-friendly interface for students, staff, and administrators. The database is used to store and manage institutional data securely, including student records, staff details, and academic information. REST APIs enable smooth interaction between system components.

These technologies are chosen to ensure efficient system performance, secure data handling, scalability, and ease of maintenance. The combination of modern development tools and frameworks makes the system reliable and suitable for real-time academic management.

# CHAPTER 4

## SYSTEM DESIGN

### 4.1 SYSTEM ARCHITECTURE



**Fig.4.1 System Architecture**

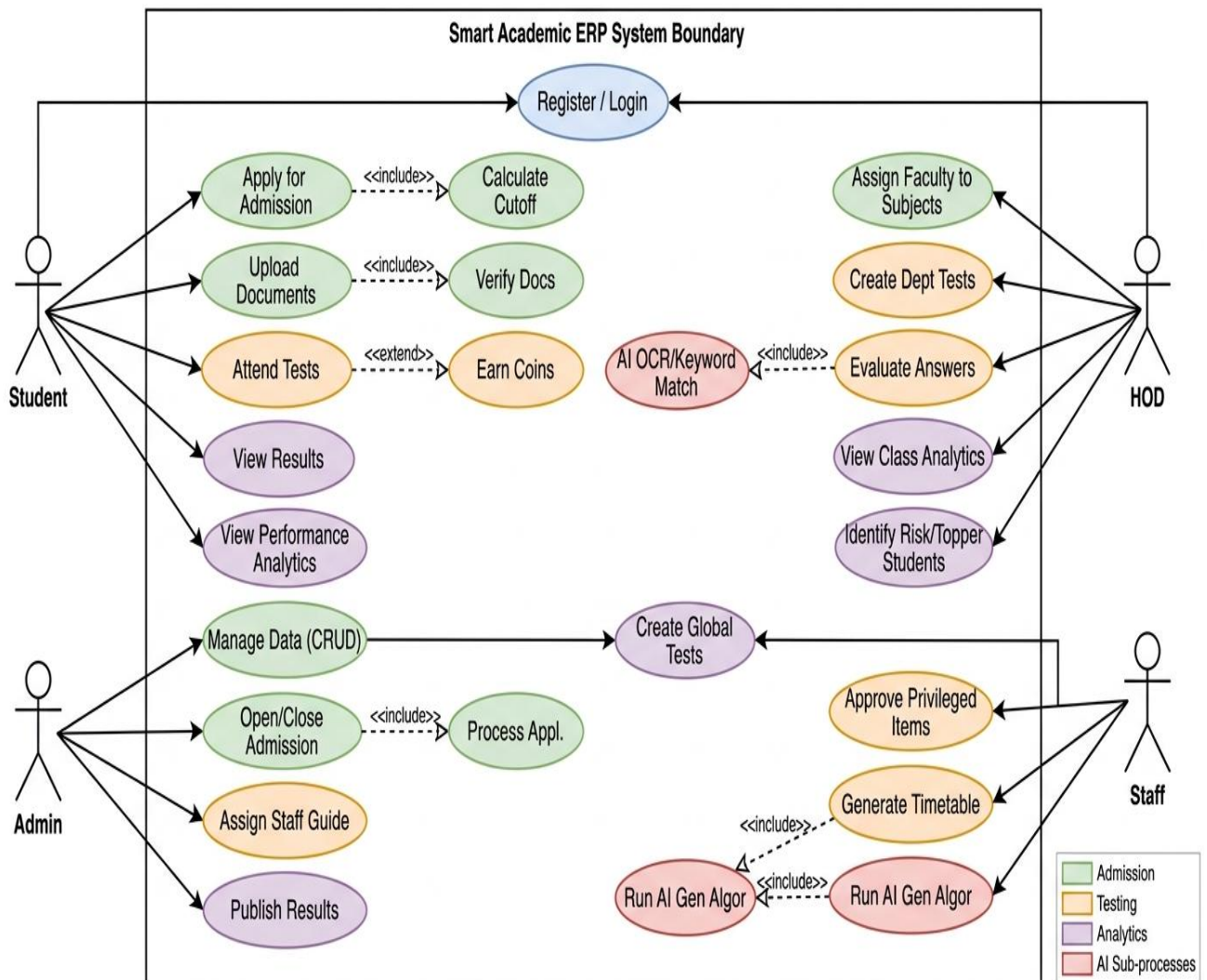
Fig. 4.1 shows the architecture of the Web-Based Smart ERP Solution, which is designed to manage academic and administrative processes in an integrated manner. The system follows a modular architecture divided into key layers such as frontend, backend,

core modules, database, and external services.

The Smart ERP System follows a multi-layered architecture designed to ensure scalability, security, and efficient management of academic and administrative processes. The system consists of three primary layers: frontend, backend, and database, along with integrated external services. The frontend layer is developed using modern web technologies and serves as the user interface through which students, staff, HODs, and administrators interact with the system. It provides functionalities such as login, registration, admission processing, test participation, and performance analysis through an intuitive and user-friendly dashboard. The frontend communicates with the backend using HTTP requests to exchange data dynamically. The backend layer, implemented using frameworks such as Spring Boot, acts as the core processing unit of the system. It handles authentication using secure mechanisms, processes user requests, executes business logic, and manages communication between different modules. The backend also interacts with the database layer, which stores structured data including student records, staff details, admission data, test results, and analytical information, ensuring data consistency, integrity, and efficient retrieval. Additionally, the system integrates external services such as OCR engines for automated evaluation of answer scripts and notification systems for sending alerts and updates to users. The overall data flow begins with user interaction at the frontend, followed by request processing in the backend, data storage and retrieval from the database, and optional interaction with external services. The processed results are then returned to the frontend and displayed to the user. This architecture ensures a modular, secure, and efficient system capable of handling complex academic workflows.

## 4.2 UML DIAGRAMS

### 4.2.1 USE CASE DIAGRAM



**Fig.4.1 Use case diagram**

Fig. 4.1 shows the use case diagram of the Smart ERP System, which represents the interaction between users and system functionalities. The system supports multiple actors such as students, staff, HODs, and administrators. Students can register, apply for admission, attend tests, and view performance. Staff can create tests and evaluate answers. HODs can assign faculty and manage timetables. Administrators control all system data. The diagram illustrates how different users interact with the system to perform various

operations.

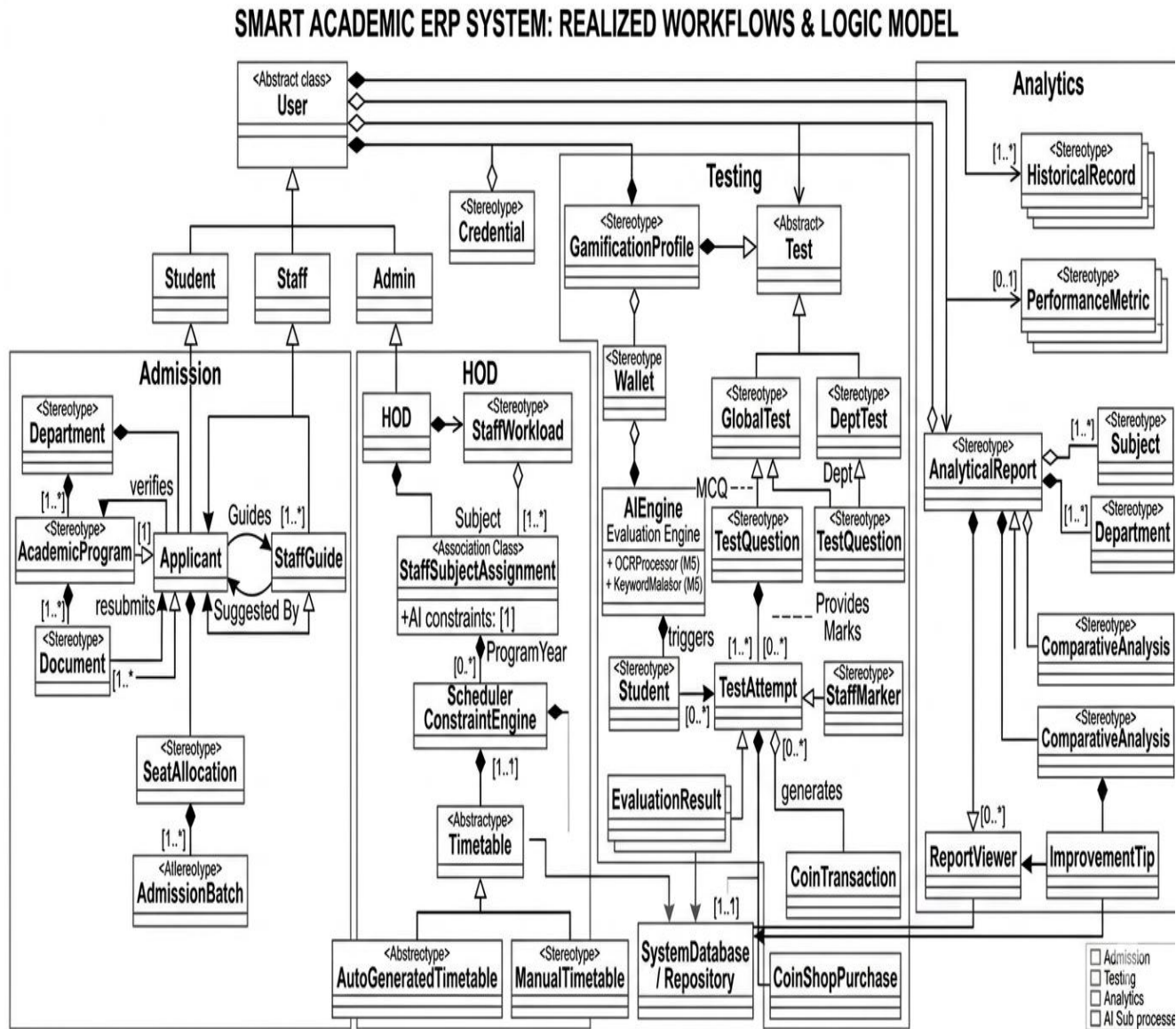
### Description:

- **Student:** The student interacts with the system to register, upload documents, apply for admission, attend tests, and view results and performance analytics.
- **Staff:** Staff members use the system to create tests and quizzes, evaluate student answers, and monitor student performance.
- **HOD (Head of Department):** The HOD is responsible for assigning faculty to subjects and generating timetables based on workload and availability.
- **Admin:** The administrator manages the entire system, including user data, academic records, and system configurations.
- **Register/Login:** Users securely log into the system using their credentials based on their roles.
- **Upload Documents:** Students upload required documents for admission verification.
- **Apply for Admission:** Students submit their application for admission through the system.
- **View Results:** Students can view their test results, evaluation outcomes, and academic performance.
- **Assign Faculty:** HOD assigns faculty members to specific subjects and classes.
- **Generate Timetable:** The system automatically generates timetables based on constraints.
- **Create Tests/Quizzes:** Staff and administrators can create assessments for students.
- **Attend Tests:** Students can participate in online tests and submit their answers.
- **Evaluate Answers (AI + Staff):** The system evaluates answers automatically, and staff can review or modify marks if needed.
- **View Performance Analytics:** Users can analyze performance through charts and

dashboards.

- **Manage Data:** Admin manages and updates all system data securely.

## 4.2.2 CLASS DIAGRAM



**Fig.4.2 Use case diagram**

Fig. 4.2 represents the class diagram of the Smart Academic ERP System, illustrating the static structure of the system. It defines the key classes, their attributes, methods, and relationships such as inheritance, association, aggregation, and composition. The diagram shows how different modules like Admission, Testing, HOD Management, and Analytics are interconnected through various classes. It provides a clear blueprint of the

system architecture and object-oriented design.

## **Description:**

- **User (Abstract Class)**

The *User* class is the base class for all types of users in the system. It contains common attributes such as user ID, name, and login credentials. It is inherited by Student, Staff, and Admin classes.

- **Student**

Represents students using the system. Students can apply for admission, attend tests, and view results and analytics. It is associated with modules like Admission, Testing, and Analytics.

- **Staff**

Staff members are responsible for creating tests, evaluating answers, and guiding students. They interact with test-related classes and evaluation processes.

- **Admin**

The Admin manages the entire ERP system. This includes controlling user data, academic programs, reports, and system configurations.

- **Credential**

Handles authentication details such as username and password. It ensures secure login functionality for all users.

## **Admission Module Classes**

- **Applicant**

Represents a student applying for admission. It includes details like submitted

documents and application status.

- **Document**

Stores uploaded documents required for admission verification.

- **Academic Program**

Defines the academic programs offered by departments.

- **Department**

Represents different departments in the institution. It verifies applicants and manages programs.

- **Seat Allocation**

Handles seat assignment for students based on eligibility and availability.

- **Admission Batch**

Represents batches of admitted students for a particular academic year.

- **Staff Guide**

Staff members who guide applicants during the admission process.

## **HOD Module Classes**

- **HOD (Head of Department)**

Responsible for managing faculty workload and subject assignments.

- **StaffSubjectAssignment**

Associates staff members with subjects and defines workload distribution.

- **StaffWorkload**

Tracks and manages workload assigned to each staff member.

- **SchedulerConstraintEngine**

Generates timetables based on constraints like availability, workload, and subjects.

- **Timetable (Abstract)**

Represents timetable structure.

- **AutoGeneratedTimetable** – Automatically generated by the system
- **ManualTimetable** – Created manually by the HOD

### **Testing Module Classes**

- **Test (Abstract Class)**

Represents a general test structure.

- **GlobalTest** – Common tests for all students
- **DeptTest** – Department-specific tests

- **TestQuestion**

Stores questions for tests (MCQ or descriptive).

- **TestAttempt**

Represents a student's attempt at a test.

- **EvaluationResult**

Stores the result of evaluated tests.

- **AIEngine**

Performs automated evaluation using OCR and keyword matching techniques.

- **StaffMarker**

Allows staff to manually evaluate or modify marks.

- **GamificationProfile**

Tracks student performance and engagement using gamification features.

- **Wallet & CoinTransaction**

Handles reward-based system where students earn coins based on performance.

- **CoinShopPurchase**

Allows students to redeem earned coins.

### **Analytics Module Classes**

- **AnalyticalReport**

Generates performance reports based on student data.

- **PerformanceMetric**

Stores performance-related metrics.

- **HistoricalRecord**

Maintains past academic records of students.

- **ComparativeAnalysis**

Compares performance across students, subjects, or departments.

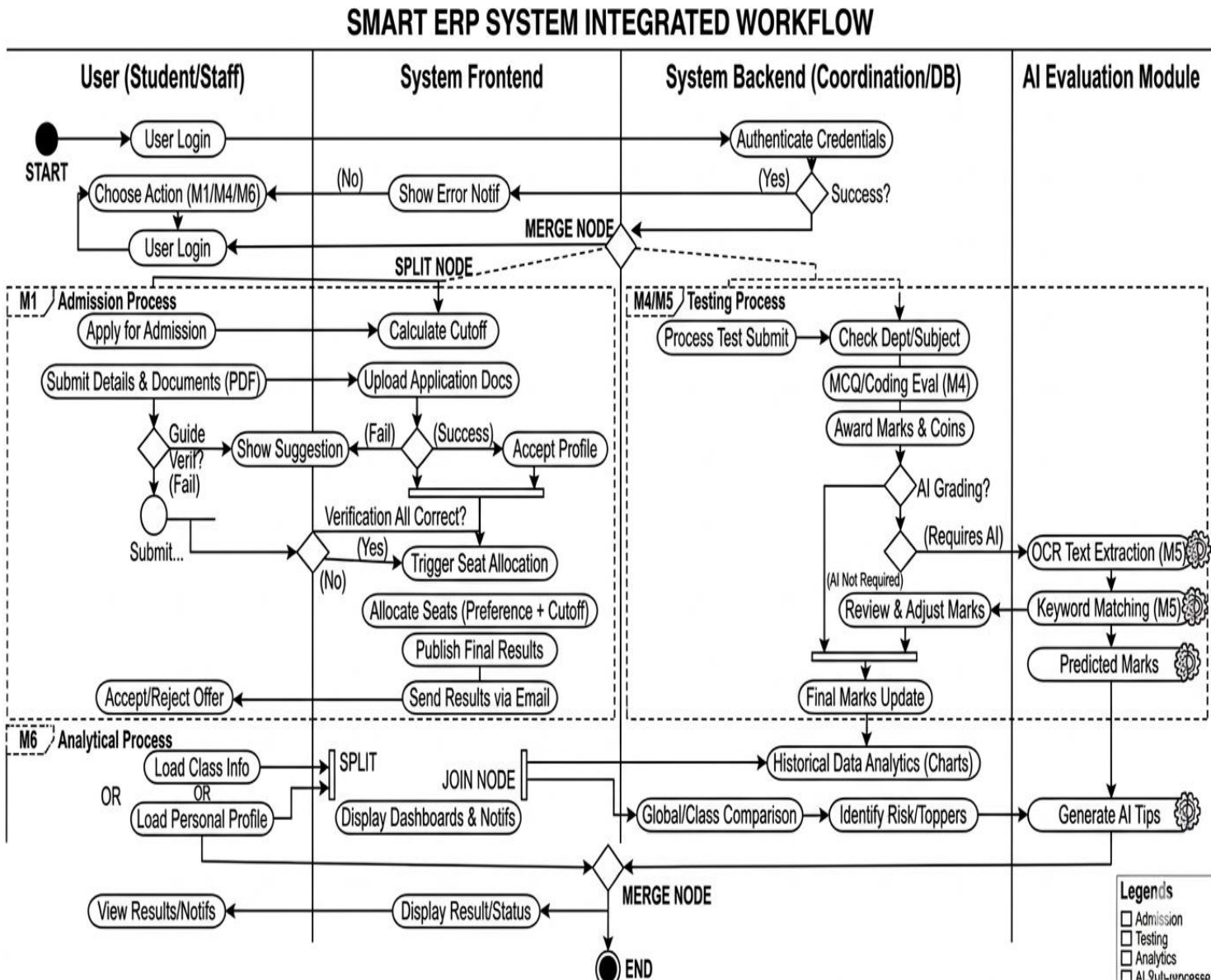
- **ImprovementTip**

Provides suggestions for improving student performance.

- **ReportViewer**

Allows users to view generated reports and analytics.

### 4.2.3 ACTIVITY DIAGRAM



**Fig.4.3 Activity Diagram**

Fig. 4.4 shows the activity diagram representing the workflow of the Smart ERP System. The process begins with user login and authentication. Based on the selected operation, the system performs actions such as admission processing, test evaluation, or performance analysis. The system processes the data, stores results in the database, and displays outputs to the user. The workflow ensures smooth execution of all academic processes.

#### **4.4.1 User Interaction**

- The process begins when the user logs into the system using valid credentials.
- The user selects the required operation such as admission, test, or analytics.
- The system provides appropriate access based on user role.

#### **4.4.2 Admission Processing**

- The student uploads required documents and submits the application.
- The system verifies documents and calculates cutoff marks.
- Seats are allocated based on merit and preferences.

#### **4.4.3 Test Processing**

- Students attend online tests through the system.
- Answers are submitted and processed by the backend.
- The system evaluates answers automatically.

#### **4.4.4 Data Processing and Analysis**

- The system processes academic data and stores it in the database.
- Performance analysis is carried out using analytical algorithms.
- Insights are generated and prepared for display.

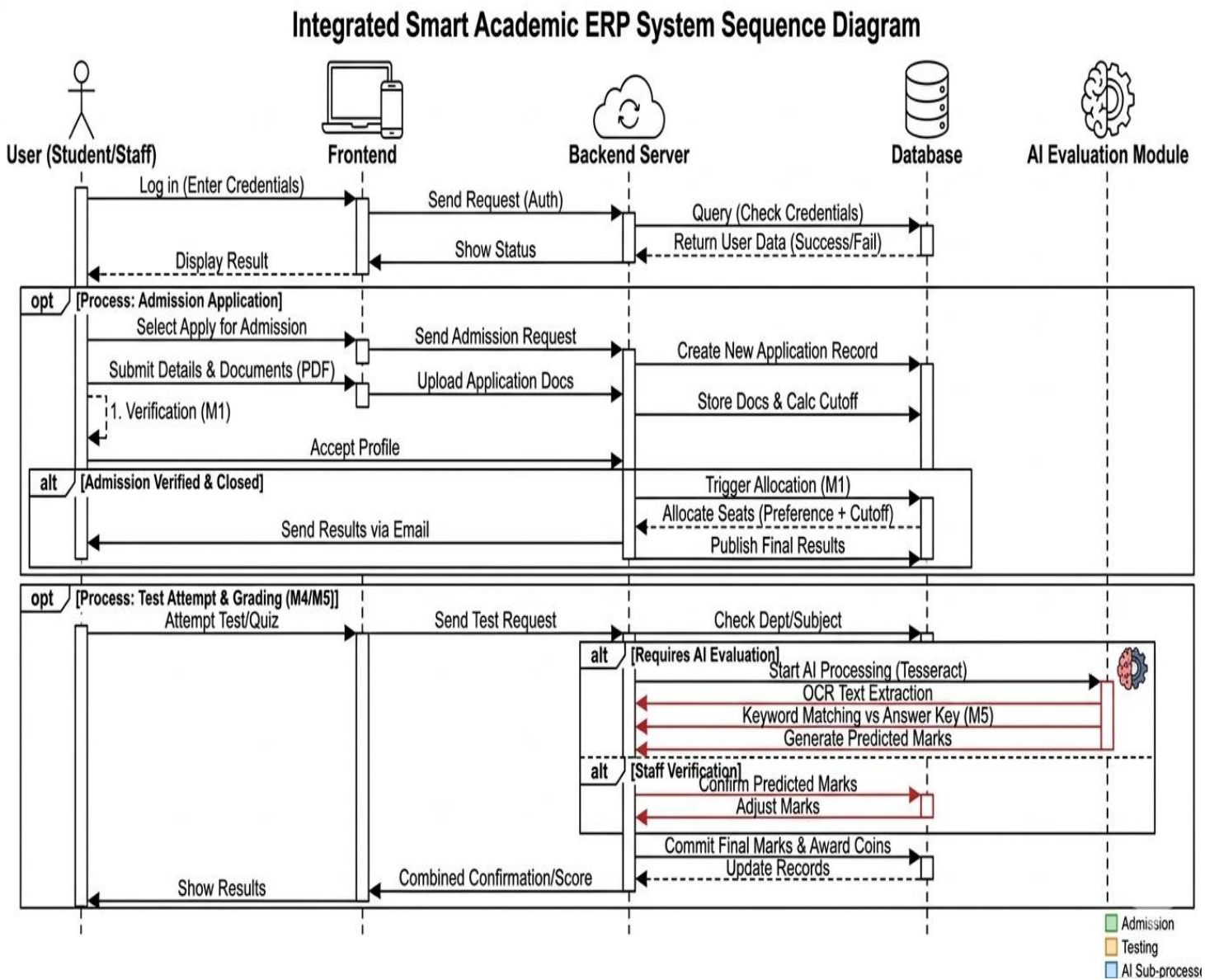
#### **4.4.5 Display Results**

- Results such as admission status, test scores, and performance analytics are displayed to the user.
- Users can view and manage their data through dashboards.

#### **4.4.6 End Process**

- The system stores all processed data securely in the database.
- Notifications are sent for important updates.
- The process continues dynamically based on user interaction

## 4.2.4 SEQUENCE DIAGRAM



**Fig.4.3 Sequence diagram**

Fig. 4.3 shows the sequence diagram representing the flow of interactions between different system components. The process begins when the user logs into the system through the frontend. The request is sent to the backend for authentication. Once verified, the user performs actions such as admission or test participation. The backend processes the request, interacts with the database, and invokes the AI evaluation module if required. The results are stored and sent back to the frontend, where they are displayed to the user.

### *Description:*

- The user (student or staff) initiates interaction with the system through the frontend interface.
- The frontend sends a request to the backend server for authentication.
- The backend verifies user credentials and grants access based on role.
- The user performs an action such as admission application, test participation, or data viewing.
- The frontend sends the request to the backend server.
- The backend processes the request and interacts with the database to fetch or store required data.
- If evaluation is required, the backend invokes the AI evaluation module to process answers.
- The AI module evaluates responses and returns results to the backend.
- The backend stores processed data in the database.
- The backend sends the response back to the frontend.
- The frontend displays the results to the user in a structured format.
- The system ensures smooth communication between all components and provides real-time update

## CHAPTER 5

### ALGORITHMS

The Smart ERP System uses a combination of algorithms to automate academic and administrative processes such as admission, timetable generation, evaluation, and performance analysis. These algorithms ensure efficient data processing, reduce manual effort, and improve accuracy. The system integrates rule-based logic, optimization techniques, and AI-based methods to perform different operations effectively.

#### 5.1 Admission Processing Algorithm

```
BEGIN Admission_Process
WHILE Admission_Status = OPEN DO
    RECEIVE student_registration_data
    VALIDATE student_data

    CALCULATE cutoff_mark for student
    STORE student_details in temporary database
END WHILE

SET Admission_Status = CLOSED

FOR EACH student IN registered_students DO
    ASSIGN mentor (AUTO or MANUAL)

    mentor_verification = VERIFY(student_documents)

    IF mentor_verification = VALID THEN
        SET student_status = VERIFIED
    ELSE
        SEND email_to_student("Correction Required")
        RECEIVE updated_data
        REPEAT verification
    END IF
END FOR
```

```

IF ALL students VERIFIED THEN
    START Seat_Allocation
END IF

SORT students BY cutoff_mark DESC

FOR EACH student IN sorted_list DO
    CHECK student_preferences

    IF seat_available THEN
        ALLOCATE seat
        SEND email("Seat Allotted")
    ELSE
        ADD to waiting_list
    END IF
END FOR

FOR EACH student DO
    IF student_accepts_offer THEN
        STORE student_data in main_database
    ELSE
        RELEASE seat
    END IF
END FOR

END

```

## 5.2 Timetable Generation Algorithm

```
BEGIN Timetable_Generation
CONNECT to database
LOAD:
    classes
    subjects (theory + lab)
    staff
    rooms
    system_config

INITIALIZE time_slots (days × periods)

CREATE empty timetable

FOR EACH subject DO
    ASSIGN required_hours_per_week
END FOR

APPLY constraints:
    - One class per slot
    - One staff per slot
    - One room per slot
    - No overlapping labs
    - No consecutive same subjects

FOR EACH subject IN subjects DO
    FIND available slots
    ASSIGN subject to slots
```

END FOR

FOR EACH lab\_subject DO

    ASSIGN consecutive slots (2 periods)

    ASSIGN lab room

END FOR

OPTIMIZE timetable:

    MINIMIZE free slots

    BALANCE workload

    REDUCE conflicts

IF solution\_found THEN

    STORE timetable in database

    GENERATE Excel and PDF

ELSE

    DISPLAY "No feasible solution"

END IF

END

### 5.3 Test Evaluation Algorithm

- BEGIN Test\_Evaluation

FOR EACH student\_submission DO

FOR EACH question DO

IF question\_type = MCQ THEN

IF answer == correct\_answer THEN

marks += full\_marks

END IF

ELSE IF question\_type = CODING THEN

EXECUTE student\_code

COMPARE output WITH expected\_output

IF match THEN

marks += full\_marks

ELSE

marks += partial\_marks

END IF

ELSE IF question\_type = THEORY THEN

MATCH keywords WITH answer

CALCULATE similarity\_score

ASSIGN marks based on score

END IF

END FOR

STORE marks in database

END FOR

END

#### 5.4 AI-Based Evaluation Algorithm

```
BEGIN AI_Evaluation
INPUT: answer_script_pdf
EXTRACT text USING OCR
FOR EACH question DO
    GET predefined_keywords
    MATCH extracted_text WITH keywords
    CALCULATE matching_score
    ASSIGN marks based on score
END FOR
DISPLAY result to staff
IF staff_modifies_marks THEN
    UPDATE marks
END IF
STORE final_marks in database
END
```

#### 5.5 Data Preprocessing

```
BEGIN Data_Preprocessing
INPUT raw_data
REMOVE null_values
REMOVE duplicate_entries
VALIDATE data_format
IF invalid_data THEN
    CORRECT or REMOVE
END IF
NORMALIZE data (format, case, structure)
STORE cleaned_data
```

END

## 5.6 Data Analysis Algorithm

```
BEGIN Data_Analysis
FETCH student_data FROM database
FOR EACH student DO
    CALCULATE:
        average_marks
        attendance_percentage
        performance_trend
END FOR
IDENTIFY:
    top_students (high marks)
    at_risk_students (low marks)
COMPARE students WITH class_average
GENERATE charts:
    - performance graph
    - comparison chart
DISPLAY results on dashboard
END
```

## 5.7 Data Storage and Management

- Processed data is stored in structured format in the database.
- It includes student records, academic data, and performance metrics.
- Ensures efficient retrieval and secure data handling.

# **CHAPTER 6**

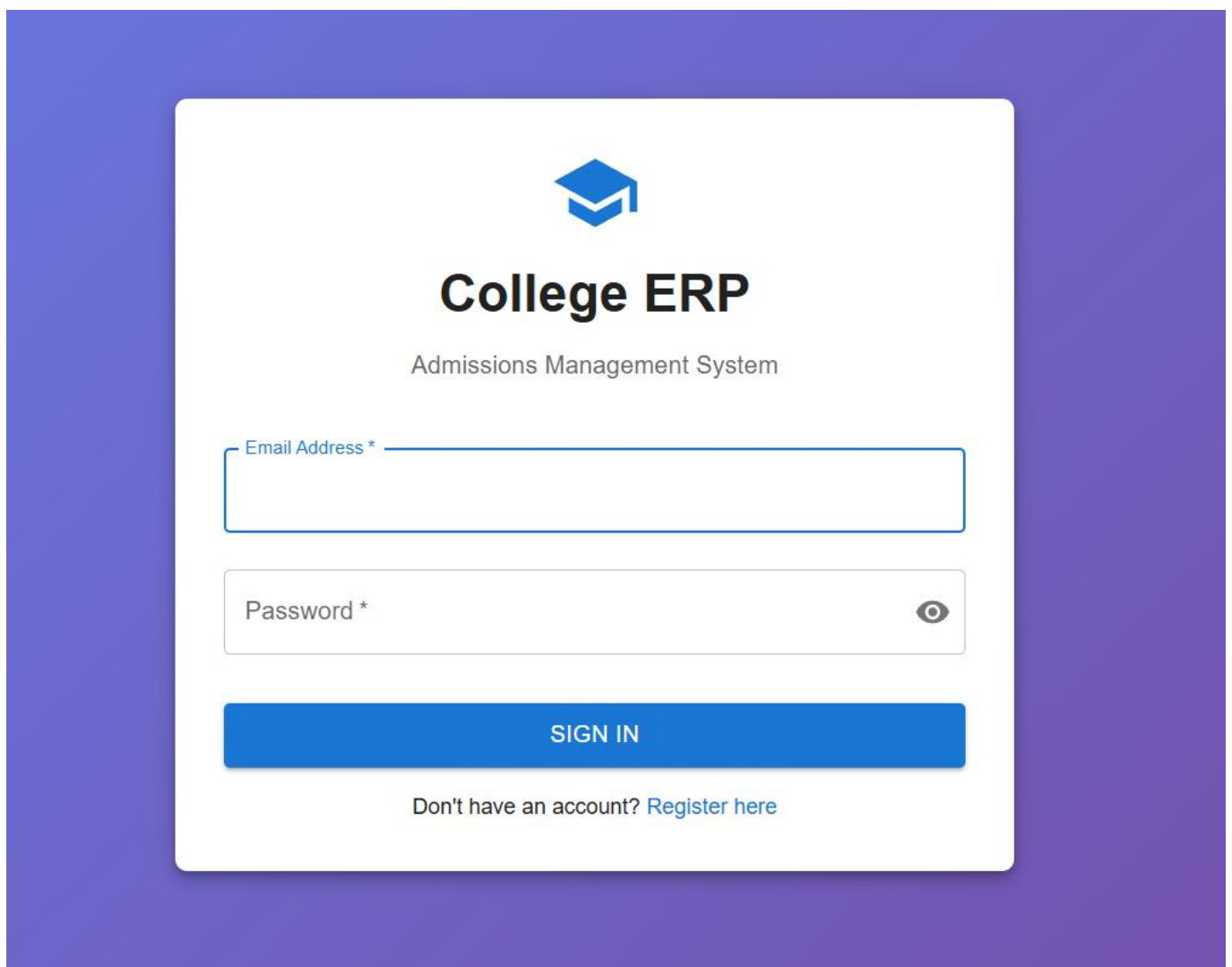
## **SYSTEM IMPLEMENTATION**

## 6.1 MODULE DESCRIPTION

### 6.1.1 AUTHENTICATION MODULE

The Authentication Module is a critical component of the Smart ERP System that ensures secure access and controlled usage of system functionalities through role-based authorization and token-based authentication mechanisms. This module is responsible for validating user credentials and managing access permissions for different types of users such as students, staff, HOD, and administrators. The system uses JSON Web Token (JWT) authentication to provide a secure and scalable login mechanism. When a user logs in, their credentials are verified against the database, and upon successful authentication, a JWT token is generated and sent to the client. This token contains encoded user information such as user ID and role, which is used for subsequent authorization without requiring repeated login verification. The role-based access control (RBAC) mechanism ensures that each user can only access the modules and functionalities permitted for their role. For example, students can access tests and performance data, staff can evaluate and create tests, HOD can manage timetables, and administrators have full system control. The module also handles session management, token validation, and secure logout operations to prevent unauthorized access. By integrating JWT and role-based authorization, the system ensures high security, scalability, and efficient access control, making it suitable for real-time web applications.

## LOGIN PAGE:



The login page features a white card centered on a purple gradient background. At the top of the card is a blue graduation cap icon. Below the icon, the text "College ERP" is displayed in a large, bold, black font, followed by "Admissions Management System" in a smaller, regular black font. The login form consists of two input fields: "Email Address \*" and "Password \*". The "Email Address \*" field is a simple white box with a blue border. The "Password \*" field is a white box with a grey border and a toggle icon (an eye) on the right side. Below the password field is a solid blue button with the text "SIGN IN" in white. At the bottom of the card, the text "Don't have an account? [Register here](#)" is displayed, with "Register here" being a blue link.



# College ERP

Admissions Management System

Email Address \*

Password \*



**SIGN IN**

Don't have an account? [Register here](#)

The above image shows the login interface of the Smart ERP System, where users enter their credentials to access the application. The interface is designed to be simple and user-friendly, allowing users to log in using their registered username and password. When the user submits the login form, the system sends the credentials to the backend server for verification. If the credentials are valid, the system generates a secure JWT token and returns it to the client. This token is then used to authenticate all subsequent requests without requiring repeated login. Based on the role associated with the user, the system redirects them to their respective dashboard, ensuring role-based access control. The login page also ensures secure handling of user data and prevents unauthorized access through proper validation and authentication mechanisms.

### **6.1.2 ADMISSION MODULE**

The Admission Module is a core component of the Smart ERP System designed to automate and streamline the entire student admission process in an efficient and transparent manner. The process begins when the administrator activates the admission cycle, allowing students to register by submitting their personal, academic, and required document details through the system. Once the data is submitted, the system automatically calculates the cutoff marks based on predefined academic criteria, ensuring fairness and consistency in evaluation. After the registration phase is closed, the administrator assigns guide staff (mentors) either manually or automatically to verify student applications. The assigned staff review the submitted documents and provide feedback or request corrections if discrepancies are found. Students are notified via email to update and resubmit their details, ensuring accuracy of information. Once all applications are verified and approved, the system initiates the seat allocation process, where seats are allotted automatically based on cutoff marks and student preferences. The final admission results are communicated to students through email, and they are given the option to accept or reject the offer. Upon acceptance, the student's data is permanently stored in the database for further academic

processing. This module reduces manual workload, eliminates errors, improves transparency, and ensures a smooth and efficient admission workflow within the institution.

## 1: Student Registration Page

The screenshot displays the 'Student Dashboard' for a student portal. The interface includes a sidebar with navigation links: Dashboard, Profile, Marks, Preferences, and Documents. The main content area features a blue header with 'College Admissions' and a user profile icon. Below the header, the 'Student Dashboard' title is followed by the 'Application Number: APP2026000005'. A light blue banner indicates the 'Registration Window: OPEN'. The 'Registration Progress' section shows a 0% completion bar and a 'SUBMIT REGISTRATION' button, with a note to 'Complete all required sections before submitting.' Below this, four status boxes are visible: 'Profile Status' (PENDING, Stream: NOT\_SELECTED, EDIT PROFILE button), 'Academic Marks' (No marks entered yet, ENTER MARKS button), 'Department Preferences' (No preferences set yet, SET PREFERENCES button), and 'Documents' (Uploaded: 0 documents, MANAGE DOCUMENTS button).

The above image shows the student registration interface where applicants enter their personal and academic details along with uploading required documents. The form is

designed to capture all necessary information such as name, contact details, previous academic scores, and supporting documents. The system validates the entered data to ensure completeness and correctness before submission. This step acts as the initial phase of the admission process and ensures that all required data is collected in a structured format. The user-friendly interface allows students to easily navigate and complete the registration process without errors.

## 2: Application List

College Admissions - Admin

System Admin S

### Student Management

Verification Status  
All Status

Students (4)

| Application No. | Name                | Email                | Stream       | Cutoff | Assigned Mentor | Verification | Registration |                 |
|-----------------|---------------------|----------------------|--------------|--------|-----------------|--------------|--------------|-----------------|
| APP2026000001   | Kaviyarasan S       | csbs259024@gmail.com | NOT_SELECTED | -      | Not assigned    | PENDING      | DRAFT        | ASSIGN / CHANGE |
| APP2026000002   | kavi                | csbs23@gmail.com     | NOT_SELECTED | -      | Not assigned    | PENDING      | DRAFT        | ASSIGN / CHANGE |
| APP2026000003   | navaneetha krishnan | csbs259023@gmail.com | NOT_SELECTED | -      | Not assigned    | PENDING      | DRAFT        | ASSIGN / CHANGE |

The above image shows the list of registered students along with their calculated cutoff marks. The system automatically processes student academic data and computes cutoff values based on predefined criteria. This ensures a fair and unbiased evaluation process. The admin can view all applications in a centralized dashboard, making it easier to manage large volumes of student data. The cutoff marks help in ranking students and play a crucial role in the seat allocation process. This automated calculation reduces manual effort and improves accuracy.

### 3: Mentor Verification Panel

Staff Portal

College Admissions - Staff

Default Staff D

Dashboard

Student Verification

## Staff Dashboard

0  
Total Assigned

0  
Pending Verification

0  
Verified

0  
Correction Requested

### Assigned Students

| Application No.          | Name | Email | Stream | Cutoff | Status |
|--------------------------|------|-------|--------|--------|--------|
| No students assigned yet |      |       |        |        |        |

The above image shows the mentor verification interface where assigned staff review student applications and documents. The mentor can check all submitted details and verify their authenticity. If any issues are found, the system allows the mentor to send feedback or correction requests to the student. Students receive notifications and can update their information accordingly. This iterative verification process ensures that only valid and accurate applications proceed to the next stage. It enhances data quality and maintains the integrity of the admission process.

## 4: Seat Allocation & Result Page

College Admissions - Admin
System Admin S

✔ No eligible students found for admissions

1 Run Admission Process
2 Review Results
3 Notify Students

▶ RUN ADMISSION PROCESS
▶ NOTIFY STUDENTS

Admission Statistics

0 Verified Students
0 Allotted
0 Not Allotted
0 Accepted Offers

Department-wise Allocation

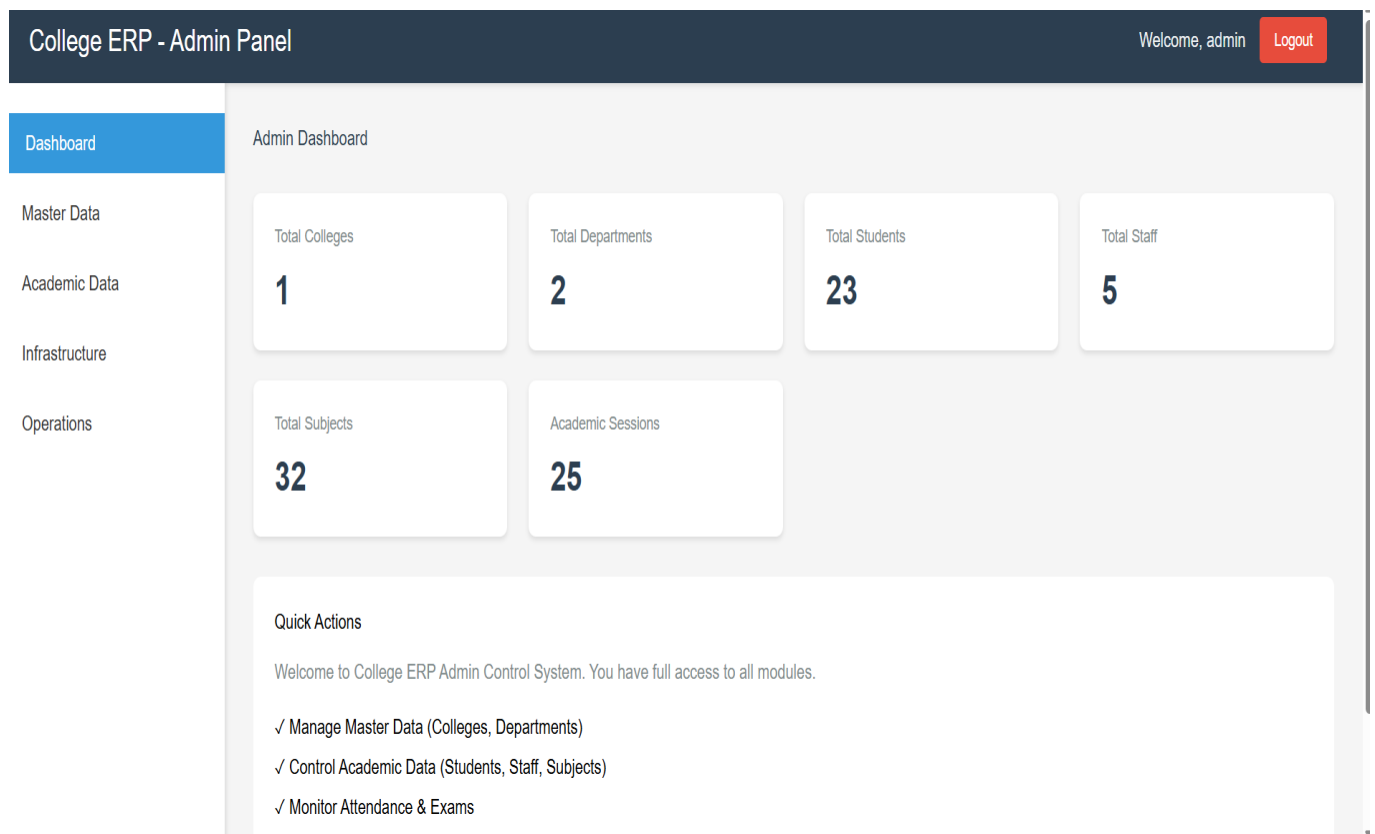
| Department                                   | Total Seats | Allocated | Available |
|----------------------------------------------|-------------|-----------|-----------|
| Computer science Engineering (CSE)           | 2           | 0         | 2         |
| Electronic communication Engineering (ECE)   | 1           | 0         | 1         |
| Computer Science and Business Systems (CSBS) | 4           | 0         | 4         |

The above image shows the seat allocation results generated by the system. Based on cutoff marks and student preferences, seats are automatically assigned in a transparent manner. The results are displayed clearly, showing the allotted program and status. Students receive the results via email and can log in to accept or reject the offer. Once accepted, the student's data is finalized and stored in the database. This automated allocation process ensures fairness, reduces bias, and improves overall efficiency of admissions.

### **6.1.3 ADMINISTRATIVE MODULE**

The Administrative Module acts as the central control unit of the Smart ERP System, responsible for managing and maintaining all institutional data in a secure and organized manner. This module provides the administrator with complete authority to monitor, update, and control various system operations, including student records, staff information, academic data, and system configurations. It ensures that all data across different modules is consistent, accurate, and up-to-date. The administrator can perform essential operations such as adding new records, updating existing data, and deleting outdated or incorrect information. This module also plays a crucial role in controlling system workflows, such as initiating or closing the admission process, managing test permissions, and overseeing system-wide activities. It ensures data integrity by enforcing validation rules and preventing unauthorized access through role-based security mechanisms. Additionally, the module provides a centralized dashboard that offers a comprehensive overview of system activities, enabling administrators to make informed decisions. By integrating all data into a single platform, this module eliminates redundancy, improves efficiency, and enhances overall system reliability. It serves as the backbone of the ERP system, ensuring smooth coordination between different modules and maintaining the overall functionality of the application.

## 1: Admin Dashboard



The above image shows the administrative dashboard, which provides a centralized view of all system activities and controls. It displays key information such as the number of

students, staff members, active admissions, and system notifications. The dashboard is designed to give administrators quick access to all modules and functionalities. It acts as the control panel where admins can monitor system status and navigate to different sections efficiently. This interface improves usability and helps administrators manage the system effectively.

## 2: Student Management Panel

| College ERP - Admin Panel             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |          |                |                                  |          |      |        |                                               |                                             |      |             |          |      |            |          |      |        |         |  |   |              |        |                |                                  |   |      |        |                                               |                                             |   |              |        |             |                                  |   |      |        |                                               |                                             |   |              |        |             |                                  |   |      |        |                                               |                                             |   |              |        |               |                                  |   |      |        |                                               |                                             |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------------|----------------------------------|----------|------|--------|-----------------------------------------------|---------------------------------------------|------|-------------|----------|------|------------|----------|------|--------|---------|--|---|--------------|--------|----------------|----------------------------------|---|------|--------|-----------------------------------------------|---------------------------------------------|---|--------------|--------|-------------|----------------------------------|---|------|--------|-----------------------------------------------|---------------------------------------------|---|--------------|--------|-------------|----------------------------------|---|------|--------|-----------------------------------------------|---------------------------------------------|---|--------------|--------|---------------|----------------------------------|---|------|--------|-----------------------------------------------|---------------------------------------------|
| Welcome, admin <a href="#">Logout</a> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |          |                |                                  |          |      |        |                                               |                                             |      |             |          |      |            |          |      |        |         |  |   |              |        |                |                                  |   |      |        |                                               |                                             |   |              |        |             |                                  |   |      |        |                                               |                                             |   |              |        |             |                                  |   |      |        |                                               |                                             |   |              |        |               |                                  |   |      |        |                                               |                                             |
| Dashboard                             | Student Management                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |          |                |                                  |          |      |        |                                               |                                             |      |             |          |      |            |          |      |        |         |  |   |              |        |                |                                  |   |      |        |                                               |                                             |   |              |        |             |                                  |   |      |        |                                               |                                             |   |              |        |             |                                  |   |      |        |                                               |                                             |   |              |        |               |                                  |   |      |        |                                               |                                             |
| Master Data                           | <a href="#">Add New Student</a> <a href="#">Show Filters ▼</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |          |                |                                  |          |      |        |                                               |                                             |      |             |          |      |            |          |      |        |         |  |   |              |        |                |                                  |   |      |        |                                               |                                             |   |              |        |             |                                  |   |      |        |                                               |                                             |   |              |        |             |                                  |   |      |        |                                               |                                             |   |              |        |               |                                  |   |      |        |                                               |                                             |
| Academic Data                         | Showing 23 of 23 students                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |          |                |                                  |          |      |        |                                               |                                             |      |             |          |      |            |          |      |        |         |  |   |              |        |                |                                  |   |      |        |                                               |                                             |   |              |        |             |                                  |   |      |        |                                               |                                             |   |              |        |             |                                  |   |      |        |                                               |                                             |   |              |        |               |                                  |   |      |        |                                               |                                             |
| Infrastructure                        | <table> <tr> <th>S.No</th><th>Register No</th><th>Batch No</th><th>Name</th><th>Department</th><th>Semester</th><th>Year</th><th>Status</th><th colspan="2">Actions</th></tr> <tr> <td>1</td><td>813818244001</td><td>189001</td><td>Aarav Krishnan</td><td>Computer Science and Engineering</td><td>8</td><td>2018</td><td>PASSED</td><td> <a href="#">Marks</a> <a href="#">Journey</a> </td><td> <a href="#">Edit</a> <a href="#">Delete</a> </td></tr> <tr> <td>2</td><td>813818244002</td><td>189002</td><td>Ananya Iyer</td><td>Computer Science and Engineering</td><td>8</td><td>2018</td><td>PASSED</td><td> <a href="#">Marks</a> <a href="#">Journey</a> </td><td> <a href="#">Edit</a> <a href="#">Delete</a> </td></tr> <tr> <td>3</td><td>813818244003</td><td>189003</td><td>Arjun Menon</td><td>Computer Science and Engineering</td><td>8</td><td>2018</td><td>PASSED</td><td> <a href="#">Marks</a> <a href="#">Journey</a> </td><td> <a href="#">Edit</a> <a href="#">Delete</a> </td></tr> <tr> <td>4</td><td>813820244004</td><td>209001</td><td>Bhavana Reddy</td><td>Computer Science and Engineering</td><td>7</td><td>2020</td><td>ACTIVE</td><td> <a href="#">Marks</a> <a href="#">Journey</a> </td><td> <a href="#">Edit</a> <a href="#">Delete</a> </td></tr> </table> |          |                |                                  |          |      |        |                                               |                                             | S.No | Register No | Batch No | Name | Department | Semester | Year | Status | Actions |  | 1 | 813818244001 | 189001 | Aarav Krishnan | Computer Science and Engineering | 8 | 2018 | PASSED | <a href="#">Marks</a> <a href="#">Journey</a> | <a href="#">Edit</a> <a href="#">Delete</a> | 2 | 813818244002 | 189002 | Ananya Iyer | Computer Science and Engineering | 8 | 2018 | PASSED | <a href="#">Marks</a> <a href="#">Journey</a> | <a href="#">Edit</a> <a href="#">Delete</a> | 3 | 813818244003 | 189003 | Arjun Menon | Computer Science and Engineering | 8 | 2018 | PASSED | <a href="#">Marks</a> <a href="#">Journey</a> | <a href="#">Edit</a> <a href="#">Delete</a> | 4 | 813820244004 | 209001 | Bhavana Reddy | Computer Science and Engineering | 7 | 2020 | ACTIVE | <a href="#">Marks</a> <a href="#">Journey</a> | <a href="#">Edit</a> <a href="#">Delete</a> |
| S.No                                  | Register No                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Batch No | Name           | Department                       | Semester | Year | Status | Actions                                       |                                             |      |             |          |      |            |          |      |        |         |  |   |              |        |                |                                  |   |      |        |                                               |                                             |   |              |        |             |                                  |   |      |        |                                               |                                             |   |              |        |             |                                  |   |      |        |                                               |                                             |   |              |        |               |                                  |   |      |        |                                               |                                             |
| 1                                     | 813818244001                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 189001   | Aarav Krishnan | Computer Science and Engineering | 8        | 2018 | PASSED | <a href="#">Marks</a> <a href="#">Journey</a> | <a href="#">Edit</a> <a href="#">Delete</a> |      |             |          |      |            |          |      |        |         |  |   |              |        |                |                                  |   |      |        |                                               |                                             |   |              |        |             |                                  |   |      |        |                                               |                                             |   |              |        |             |                                  |   |      |        |                                               |                                             |   |              |        |               |                                  |   |      |        |                                               |                                             |
| 2                                     | 813818244002                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 189002   | Ananya Iyer    | Computer Science and Engineering | 8        | 2018 | PASSED | <a href="#">Marks</a> <a href="#">Journey</a> | <a href="#">Edit</a> <a href="#">Delete</a> |      |             |          |      |            |          |      |        |         |  |   |              |        |                |                                  |   |      |        |                                               |                                             |   |              |        |             |                                  |   |      |        |                                               |                                             |   |              |        |             |                                  |   |      |        |                                               |                                             |   |              |        |               |                                  |   |      |        |                                               |                                             |
| 3                                     | 813818244003                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 189003   | Arjun Menon    | Computer Science and Engineering | 8        | 2018 | PASSED | <a href="#">Marks</a> <a href="#">Journey</a> | <a href="#">Edit</a> <a href="#">Delete</a> |      |             |          |      |            |          |      |        |         |  |   |              |        |                |                                  |   |      |        |                                               |                                             |   |              |        |             |                                  |   |      |        |                                               |                                             |   |              |        |             |                                  |   |      |        |                                               |                                             |   |              |        |               |                                  |   |      |        |                                               |                                             |
| 4                                     | 813820244004                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 209001   | Bhavana Reddy  | Computer Science and Engineering | 7        | 2020 | ACTIVE | <a href="#">Marks</a> <a href="#">Journey</a> | <a href="#">Edit</a> <a href="#">Delete</a> |      |             |          |      |            |          |      |        |         |  |   |              |        |                |                                  |   |      |        |                                               |                                             |   |              |        |             |                                  |   |      |        |                                               |                                             |   |              |        |             |                                  |   |      |        |                                               |                                             |   |              |        |               |                                  |   |      |        |                                               |                                             |
| Operations                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |          |                |                                  |          |      |        |                                               |                                             |      |             |          |      |            |          |      |        |         |  |   |              |        |                |                                  |   |      |        |                                               |                                             |   |              |        |             |                                  |   |      |        |                                               |                                             |   |              |        |             |                                  |   |      |        |                                               |                                             |   |              |        |               |                                  |   |      |        |                                               |                                             |

The above image shows the student management interface where the administrator can view and manage all student records. It includes functionalities such as adding new students, updating existing records, and deleting unnecessary data. The system ensures that

all student information is stored in a structured format, making it easy to retrieve and modify. This panel plays a vital role in maintaining accurate student data and supporting other modules such as admission and analytics.

### 3: Staff Management Panel

| College ERP - Admin Panel |                  |                    |                                  |                     |                           |            |        | Welcome, admin | Logout |
|---------------------------|------------------|--------------------|----------------------------------|---------------------|---------------------------|------------|--------|----------------|--------|
| Dashboard                 | Staff Management |                    |                                  |                     |                           |            |        |                |        |
| Master Data               | Add New Staff    |                    |                                  |                     |                           |            |        |                |        |
| Academic Data             |                  |                    |                                  |                     |                           |            |        |                |        |
| Infrastructure            |                  |                    |                                  |                     |                           |            |        |                |        |
| Operations                |                  |                    |                                  |                     |                           |            |        |                |        |
| ID                        | Employee ID      | Name               | Department                       | Designation         | Qualification             | Experience | Status | Actions        |        |
| 1                         | SVCE001          | Dr. Rajesh Kumar   | Computer Science and Engineering | HOD                 | Ph.D. in Computer Science | 15 years   | ACTIVE | Edit           | Delete |
| 2                         | SVCE002          | Dr. Priya Sharma   | Computer Science and Engineering | Associate Professor | Ph.D. in AI/ML            | 12 years   | ACTIVE | Edit           | Delete |
| 3                         | SVCE003          | Mr. Karthik Venkat | Computer Science and Engineering | Assistant Professor | M.E. Computer Science     | 8 years    | ACTIVE | Edit           | Delete |
|                           |                  |                    |                                  |                     |                           |            |        |                |        |

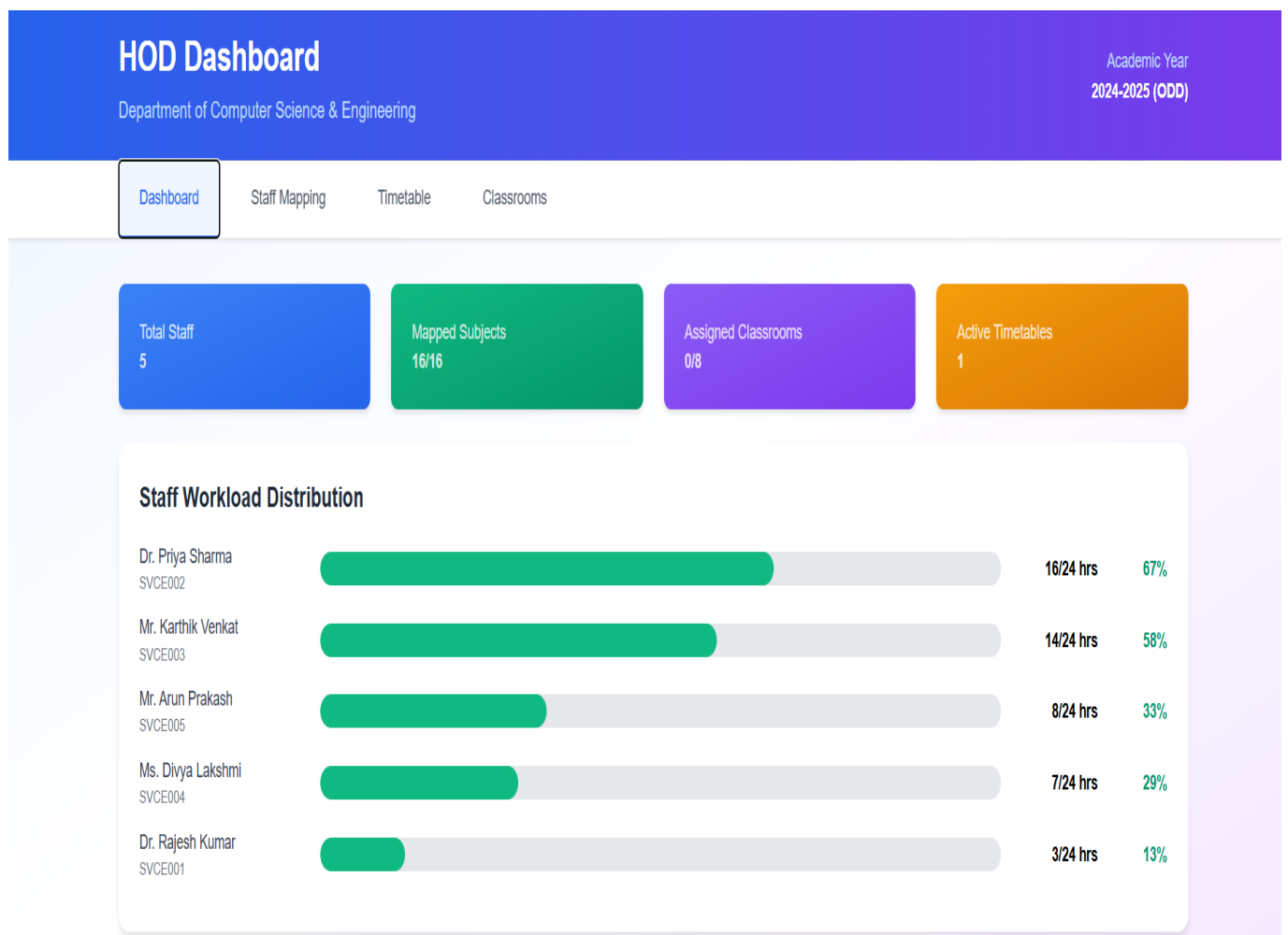
The above image shows the staff management section, where administrators can manage faculty details and their roles. It allows adding new staff members, assigning roles,

updating information, and removing inactive users. This ensures that only authorized staff members are part of the system. Proper staff management is essential for modules like HOD assignment, timetable generation, and evaluation processes. The interface is designed to simplify staff administration tasks.

#### **6.1.4 HOD MODULE**

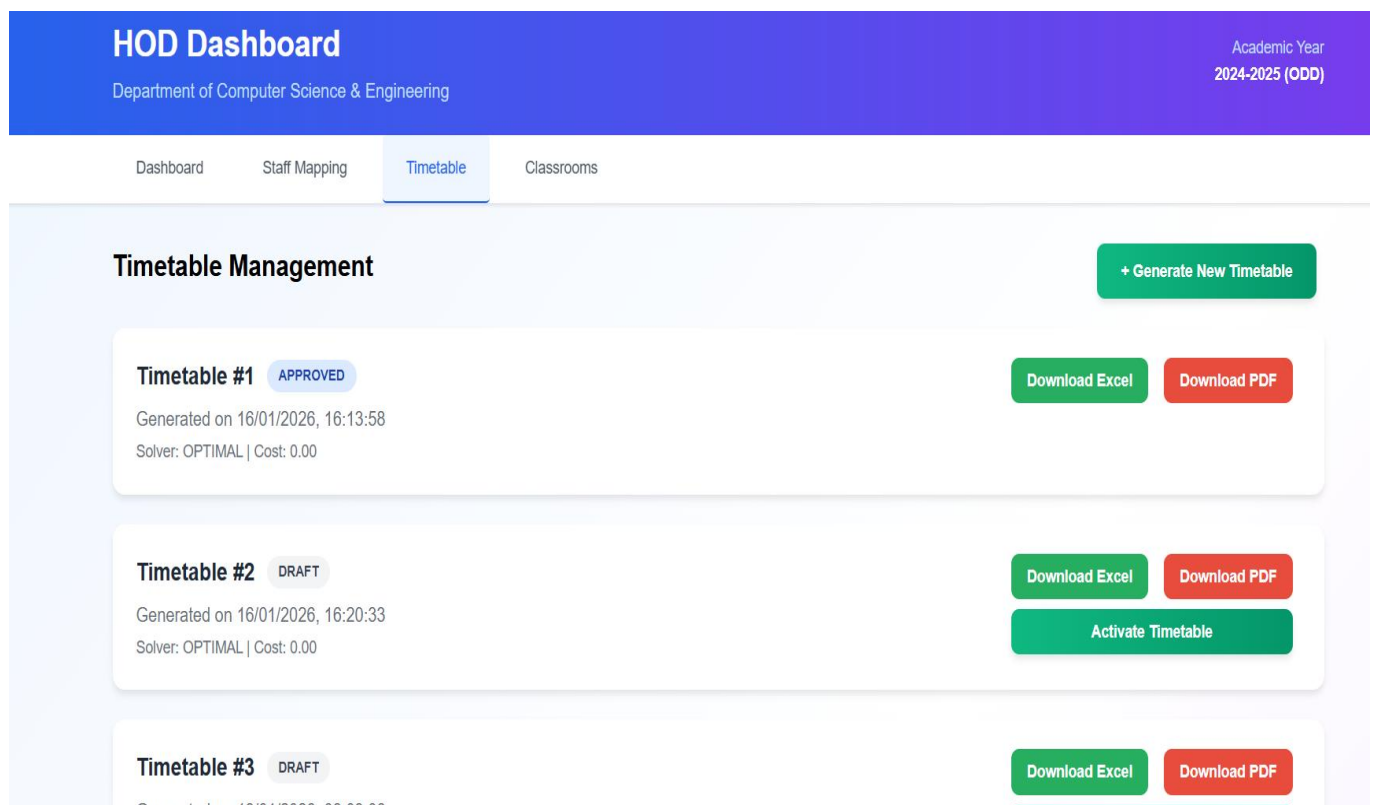
The HOD (Head of Department) Module is designed to manage and coordinate academic activities within a specific department efficiently. This module provides the HOD with control over faculty assignment and timetable generation, ensuring proper academic planning and resource utilization. The HOD can assign staff members to specific subjects and classes based on their expertise, availability, and workload constraints. The system maintains a balanced distribution of workload among faculty members to avoid overloading or underutilization. One of the key features of this module is the automatic timetable generation, where the system uses scheduling algorithms to create conflict-free timetables by considering various constraints such as staff availability, subject requirements, and time slots. The generated timetable ensures that no faculty or class is assigned multiple tasks at the same time. Additionally, the HOD can manually adjust the timetable if required, providing flexibility in academic planning. The system also supports exporting timetables in formats such as PDF or Excel, making it easy to share and use. All generated timetables are stored in the database for future reference. This module significantly reduces manual effort, improves scheduling efficiency, and ensures smooth academic operations within the department.

## 1: Faculty Assignment Page



The above image shows the faculty assignment interface where the HOD assigns staff members to specific subjects and classes. The system displays available staff along with their workload, helping the HOD make informed decisions. The assignment process ensures that subjects are allocated based on staff expertise and availability. This feature helps in maintaining a balanced workload among faculty members and avoids scheduling conflicts in later stages. It also provides flexibility for manual adjustments when required.

## 2: Timetable Generation Page



The above image shows the timetable generation interface where the HOD initiates the automatic scheduling process. The system uses predefined algorithms to generate a

conflict-free timetable by considering constraints such as staff availability, subject distribution, and time slots. Once generated, the timetable is displayed in a structured format, making it easy to understand and verify. This automation significantly reduces manual scheduling effort and improves accuracy.

### **6.1.5 CONNECTION PLATFORM MODULE**

The Connection Platform Module is designed to enhance interaction and engagement between students, staff, and administrators through an integrated digital learning and assessment environment. This module allows the admin and HOD to create globally accessible tests and quizzes, while staff members can create department-specific assessments for their students. The system supports multiple types of questions, including multiple-choice questions (MCQs) and coding-based problems, enabling a comprehensive evaluation approach. Students can participate in these tests through an online interface and submit their answers in real time. The system automatically evaluates objective questions and coding outputs, providing instant feedback and results. A unique feature of this module is the reward-based coin system, where students earn coins based on their performance in tests and activities. These coins can be used to unlock academic benefits such as representation positions, on-duty permissions, or other institutional privileges. Certain high-value rewards require approval from the HOD, ensuring controlled access. This module not only improves academic assessment but also motivates students to actively participate and perform better. It creates an interactive learning environment, bridges the gap between students and faculty, and promotes continuous engagement within the academic system.

## 1: Test / Quiz Creation Page

College Connect kavi Logout

- Dashboard
- Contests
- Create Contest**
- Wallet
- Store
- Leaderboard
- Admin
- Store Management

### Create New Contest

1 Contest Details — 2 MCQ Questions — 3 Code Problems — 4 Review

**Contest Details**

Title \*

Description

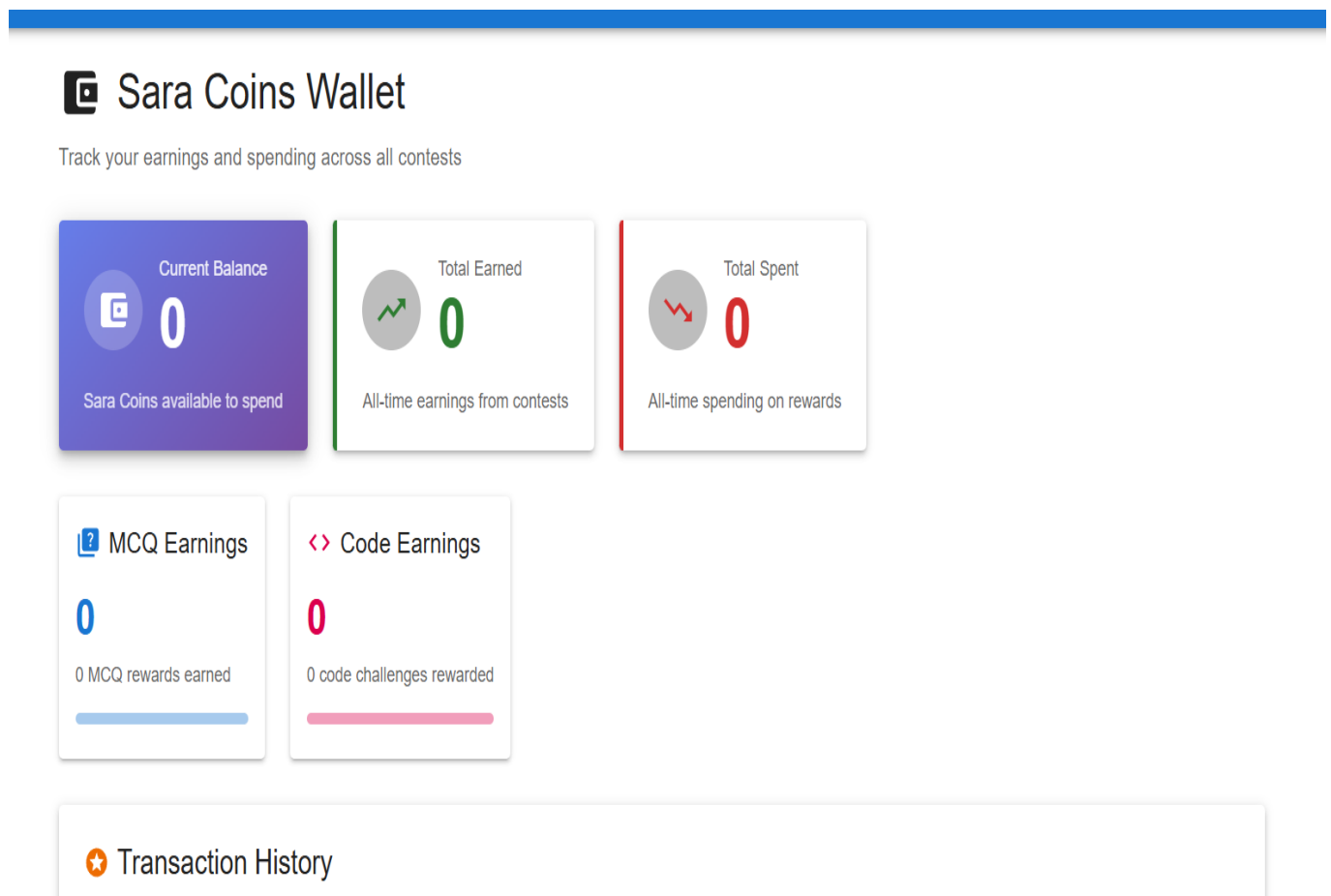
Scope \*  
Department

Department Code  
CSE

The above image shows the test creation interface where admin or staff can create quizzes and coding tests. The system allows adding different types of questions such as MCQs and programming problems. The creator can define question details, correct answers,

and evaluation criteria. This flexible interface supports customized test creation for different subjects and departments. It ensures that assessments are structured and aligned with academic requirements.

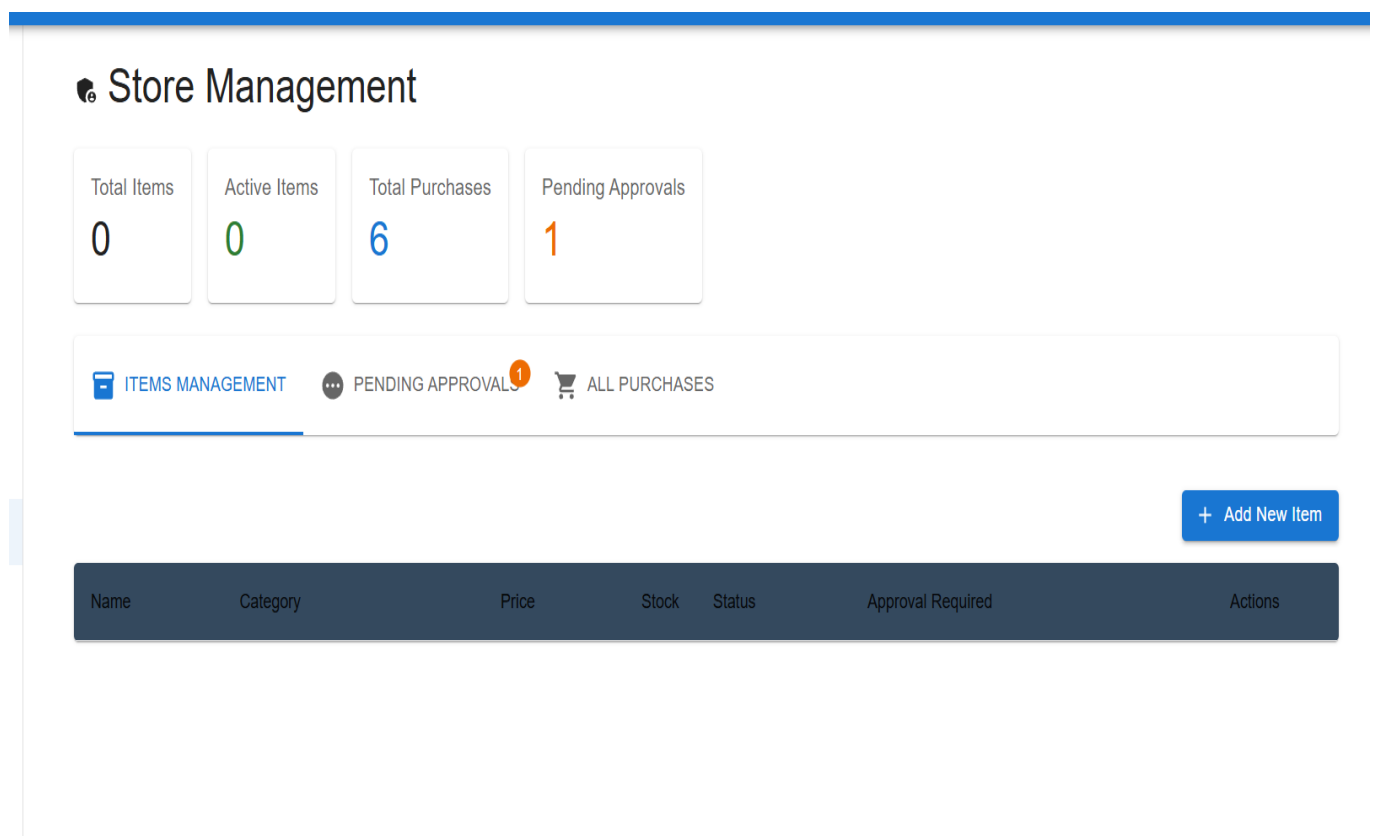
## 2: Result & Coin Dashboard



The above image shows the result and reward dashboard where students can view their test scores and earned coins. The system calculates marks automatically and updates

the student's coin balance based on performance. The dashboard displays accumulated coins and available rewards that can be redeemed. This feature motivates students to participate actively and improve their performance. It also introduces a gamified learning experience within the academic system.

### 3: Reward / Approval Page (HOD Control)



The above image shows the reward management interface where students can redeem coins for academic benefits. Some rewards require approval from the HOD, ensuring

controlled access to privileges. The HOD can review requests and approve or reject them based on institutional policies. This ensures fairness and prevents misuse of the reward system. It adds an extra layer of control while maintaining student motivation.

### **6.1.6: EVALUATION MODULE**

The AI Evaluation Module is designed to automate the evaluation of descriptive answer scripts using intelligent techniques such as Optical Character Recognition (OCR) and keyword-based analysis. This module significantly reduces the manual effort involved in grading and ensures consistency in evaluation. The process begins when staff members upload answer keys along with predefined keywords for a specific exam using an exam ID. The system then retrieves the list of students who have attended the exam and allows staff to upload scanned answer scripts in PDF format. Once uploaded, the system uses OCR technology to extract textual content from the documents. The extracted text is then processed and compared with the predefined keywords and reference answers using matching algorithms. Based on the level of similarity and keyword presence, marks are automatically assigned to each answer. The evaluated results are displayed to the staff for review, and they are given the flexibility to modify marks if necessary. This hybrid approach ensures both automation and accuracy. The final marks are then stored in the database, and the system also provides a dashboard showing evaluation insights, predictions, and performance trends. This module enhances efficiency, reduces evaluation time, and introduces intelligent automation into the academic assessment process.

## 1: Answer Key & Keyword Entry Page

AI Answer Sheet Evaluation

[Home](#) [Admin](#) [Answer Keys](#) [Evaluation](#) [Statistics](#)

### Answer Key Management

**Select Exam & Subject**

Exam ID \*

Enter Exam ID

Subject ID \*

Enter Subject ID

Load Answer Keys

**Create Answer Key**

Question Number \*

1

Max Marks \*

10.00

Question Text (Optional)

Enter the question text for reference

Marking Scheme (Optional)

Detailed marking scheme or rubric

The above image shows the answer script upload interface where staff upload student answer sheets in PDF format. The system also displays the list of students who attended the exam, allowing easy mapping of scripts to students. This ensures that all submissions are

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processed correctly. The upload feature is designed to handle multiple files efficiently and initiate the evaluation process.

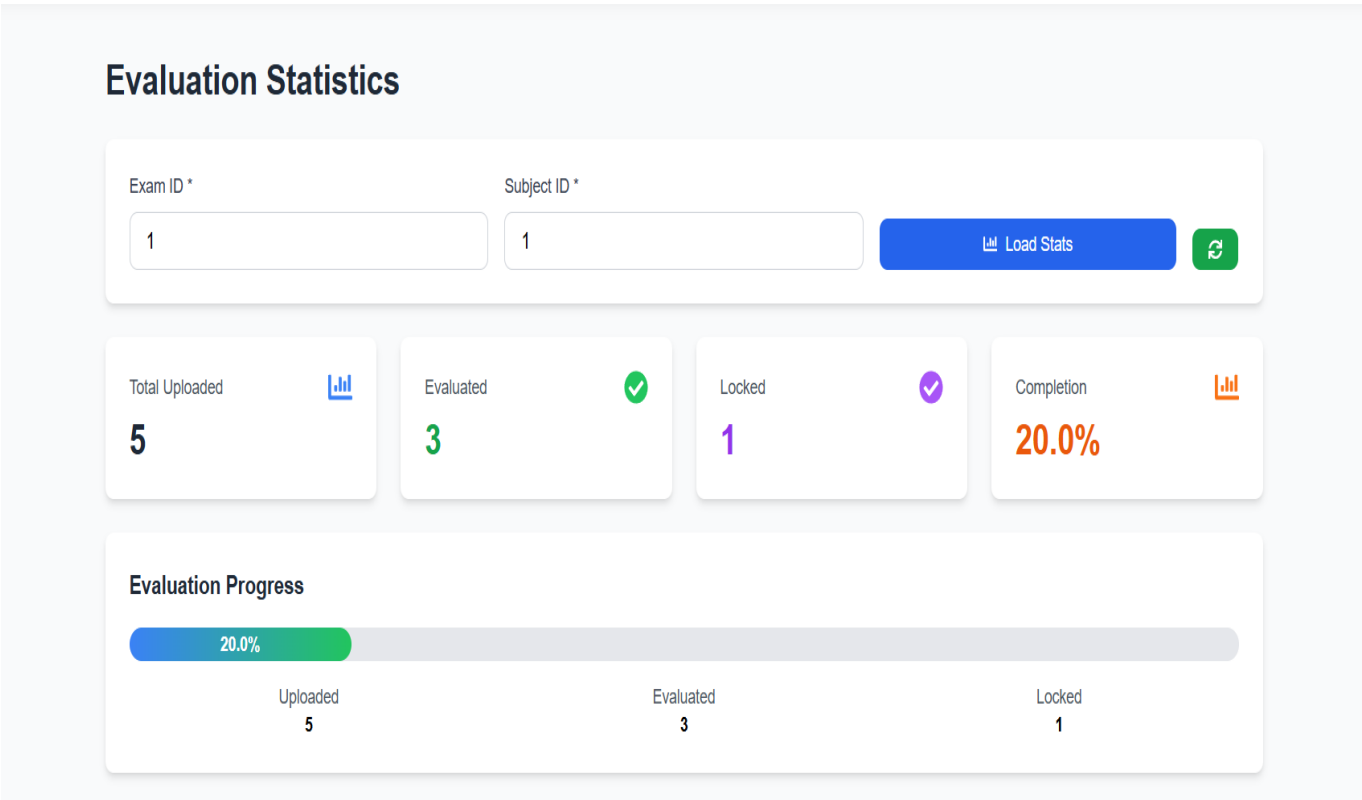
## 2: Answer Script Upload Page

The screenshot displays the 'Answer Sheet Evaluation' interface. It features a white card with a light gray border. At the top of the card, the title 'Answer Sheet Evaluation' is written in a bold, dark blue font. Below the title, the section 'Enter Exam Details' is centered. Under this section, there are three input fields arranged horizontally. The first field is labeled 'Exam ID \*' and contains the number '1'. The second field is labeled 'Subject ID \*' and also contains the number '1'. The third field is labeled 'Session ID \*' and contains the number '25', with a small upward and downward arrow icon on its right side. Below these three fields is a wide, solid blue button with the text 'Load Students' in white, centered. At the bottom of the page, below the card, there is a small copyright notice: '© 2026 AI Answer Sheet Evaluation System. All rights reserved.'

The above image shows the answer script upload interface where staff upload student answer sheets in PDF format. The system also displays the list of students who attended the exam, allowing easy mapping of scripts to students. This ensures that all submissions are

processed correctly. The upload feature is designed to handle multiple files efficiently and initiate the evaluation process.

### 3: Evaluation Analytics Dashboard



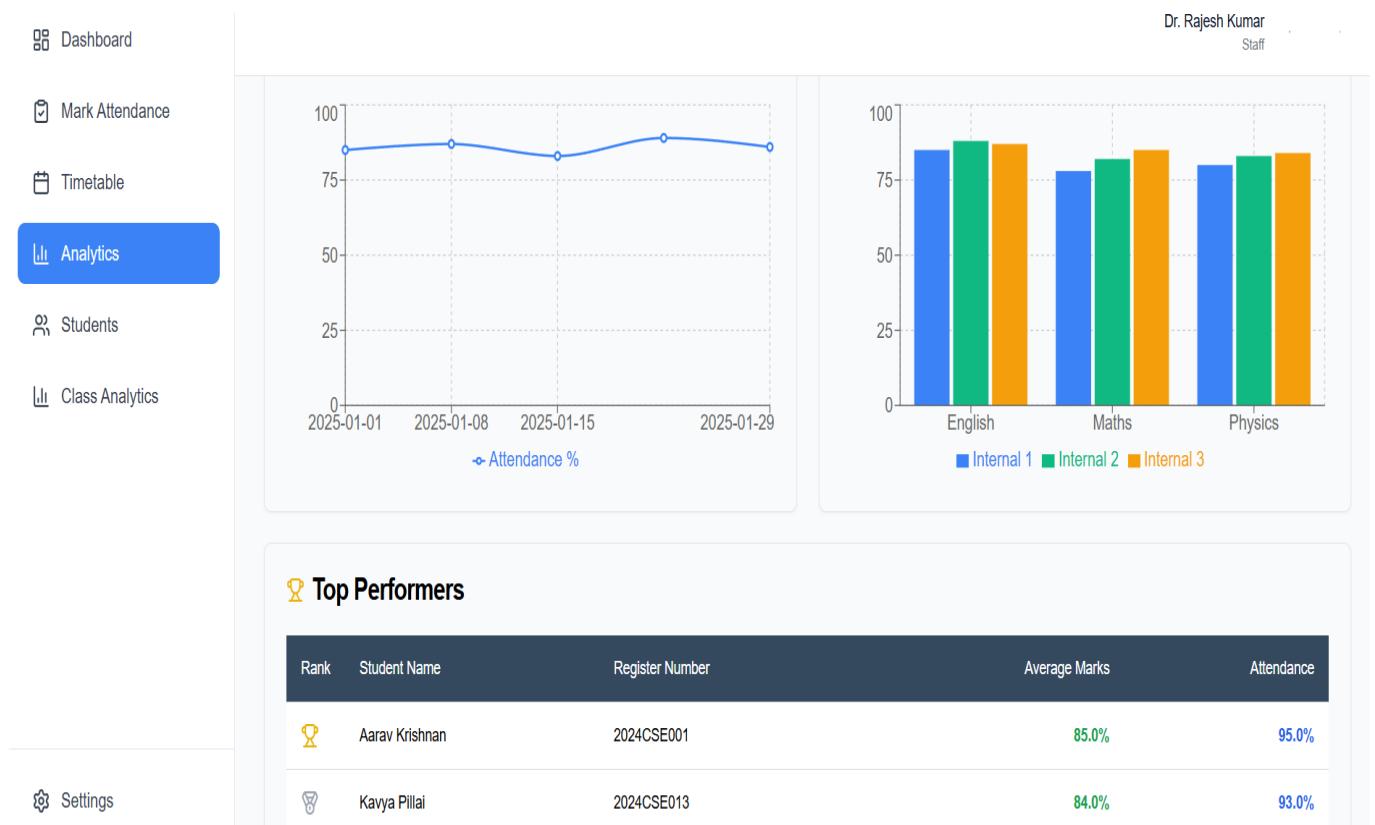
The above image shows the analytics dashboard that provides insights into the evaluation process. It displays metrics such as average marks, distribution of scores, and

performance trends. This helps staff understand overall student performance and identify patterns. The dashboard enhances decision-making and provides a clear overview of evaluation outcomes.

### **6.1.7 ANALYTICAL MODULE**

The Analytical Module is designed to provide meaningful insights into student performance and academic progress using data visualization and analysis techniques. This module plays a crucial role in transforming raw academic data into actionable information for both staff and students. Staff members can load and analyze student data for their respective classes, allowing them to track performance trends over time. The system generates various charts and reports that help in identifying high-performing students as well as those who are at risk. It enables comparative analysis between students, subjects, and overall class performance, supporting data-driven decision-making. Students also have access to their individual performance dashboards, where they can view their academic progress, compare their results with peers, and understand their strengths and weaknesses. The module includes a notification feature that alerts students about risks, important events, and performance issues. Additionally, the system provides improvement suggestions based on performance patterns, helping students enhance their academic outcomes. By offering a clear and visual representation of data, this module improves transparency, supports continuous monitoring, and enables proactive academic management within the institution.

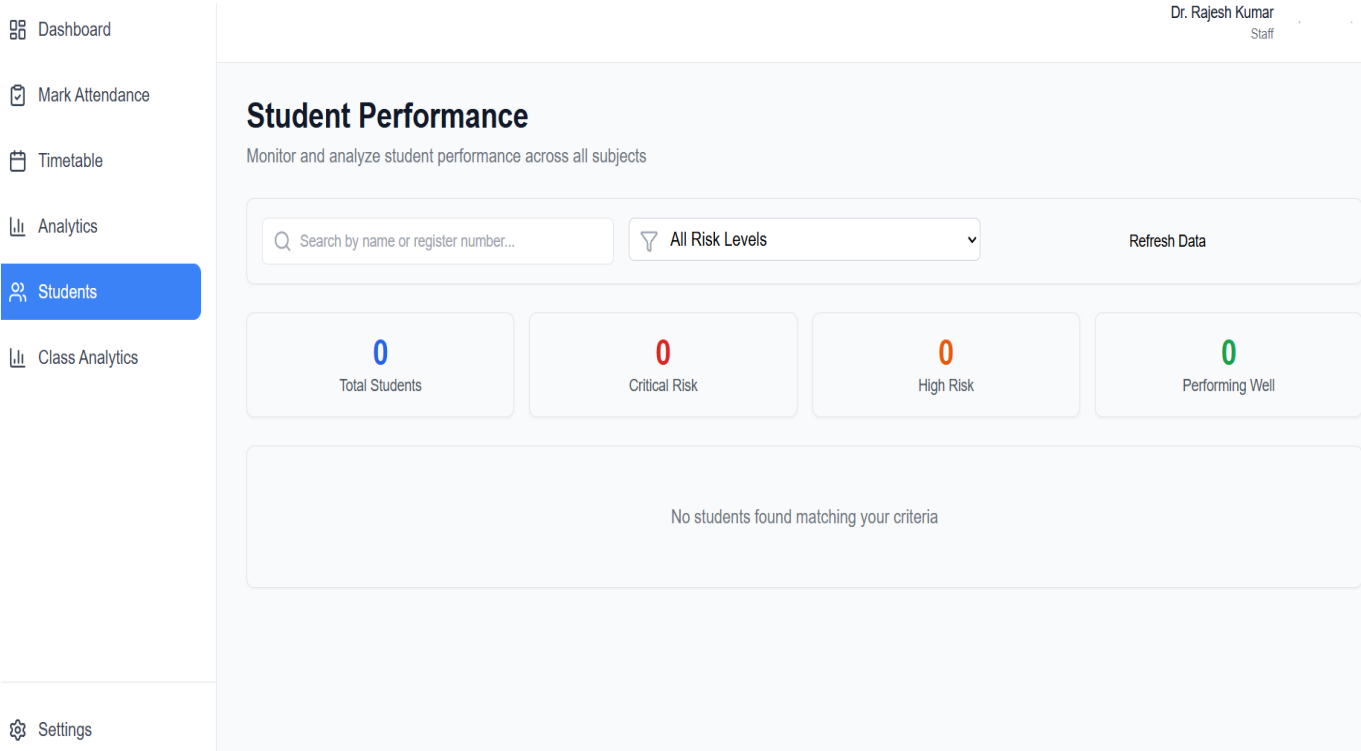
## 1: Student Performance View



The above image shows the individual student performance page where students can track their academic progress. It displays marks, test results, and performance trends over

time. Students can easily understand their strengths and areas that need improvement. This helps them focus on weak areas and improve their overall performance. The interface is designed to be simple and informative.

## 4: Risk Identification & Alerts



The above image shows the risk identification feature where the system highlights

students who are underperforming. It generates alerts and notifications for both students and staff. This helps in early identification of academic issues and allows timely intervention. The system ensures that no student is overlooked and supports proactive academic management.

## CHAPTER 7

### TOOLS AND TECHNOLOGIES

#### 7.1 SOFTWARE DESCRIPTION

##### Frontend Technologies

###### **React.js:**

- **Definition:** A JavaScript library used for building dynamic and interactive user interfaces.
- **Implementation:** React is used to develop the frontend of the ERP system, including dashboards for students, staff, HODs, and administrators. It enables component-based architecture, improves performance using virtual DOM, and provides a responsive user experience.

###### **HTML5:**

- **Definition:** Standard markup language used to structure web content.
- **Implementation:** Used to design the basic structure of the application such as forms, pages, and layouts for different modules like admission and analytics.

### CSS3:

- **Definition:** Style sheet language used for designing and layout.
- **Implementation:** Provides styling for the user interface, including layout design, colors, responsiveness, and visual enhancements for dashboards and pages.

### JavaScript:

- **Definition:** Client-side scripting language used for dynamic interaction.
- **Implementation:** Handles user interactions such as form submissions, API calls, navigation, and real-time updates in the application.

## Backend Technologies

### Java:

- **Definition:** A high-level, object-oriented programming language widely used for enterprise applications.
- **Implementation:** Used to develop the backend logic of the ERP system, including handling requests, processing data, and managing business logic.

### Spring Boot:

- **Definition:** A Java-based framework used for building scalable and production-ready web applications.
- **Implementation:** Used to create REST APIs, manage backend services, handle

authentication, and connect the frontend with the database. It simplifies development through dependency management and built-in configurations.

## Data Storage

### **MySQL:**

- **Definition:** A relational database management system used for storing structured data.
- **Implementation:** Used to store and manage data such as student records, staff details, admission data, test results, and performance analytics. It ensures data integrity, consistency, and efficient querying.

## APIs and Communication

### REST APIs:

- **Definition:** Architectural style used for communication between frontend and backend.
- **Implementation:** Used to send and receive data between React frontend and Spring Boot backend, ensuring smooth interaction between system components.

## Authentication

### Role-Based Authentication:

- **Definition:** Security mechanism that restricts system access based on user roles.
- **Implementation:** Used to provide different access levels for students, staff, HODs, and administrators, ensuring secure and controlled system usage.

## Development Tools

### Visual Studio Code:

- **Definition:** A lightweight code editor with debugging support.
- **Implementation:** Used for developing frontend and backend code, managing files, and debugging applications.

### Postman:

- **Definition:** API testing tool.
- **Implementation:** Used to test REST APIs, verify request-response behavior, and ensure proper backend functionality.

### Git & GitHub:

- **Definition:** Version control system for tracking code changes.
- **Implementation:** Used for maintaining code history, collaboration, and project management.

## 7.2 SOFTWARE SPECIFICATIONS

**Table 6.1 Software Specification**

| Component             | Specification                |
|-----------------------|------------------------------|
| Operating System      | Windows 10/11                |
| Programming Language  | java                         |
| Framework             | Spring Boot                  |
| Frontend Technologies | React, HTML, CSS, JavaScript |
| Database              | MySQL                        |
| APIs                  | REST APIs                    |
| Authentication        | Role-Based Authentication    |
| Code Editor           | VS Code                      |

## 7.2 HARDWARE DESCRIPTION

**Table 6.2 Hardware Specification**

| Component           | Specification                        |
|---------------------|--------------------------------------|
| Processor           | Intel Core i3 / i5 or equivalent     |
| RAM                 | 4 GB (8 GB recommended)              |
| Storage             | 50 GB (SSD preferred)                |
| Internet Connection | Stable network (5–10 Mbps or higher) |

Graphics

Not required

## CHAPTER 8

### TESTING

Testing plays a crucial role in ensuring the accuracy, reliability, and overall performance of the Web-Based Smart ERP System. The system is tested using both black box and white box testing techniques to validate functional correctness as well as internal logic. These testing approaches help in identifying errors, improving system performance, and ensuring that all modules operate as expected. The testing process covers various aspects such as functionality, integration, security, and performance to ensure a robust and reliable system.

#### 8.1 BLACK BOX TESTING

Black box testing focuses on validating the system functionality without considering the internal implementation. The system is tested based on inputs and expected outputs, ensuring that all features work correctly from the user's perspective. This type of testing is essential for verifying real-world scenarios and user interactions.

##### Types of Black Box Testing Used:

- **Functional Testing:** Ensures that each module such as admission, test, evaluation, and analytics performs its intended function correctly.
- **Integration Testing:** Verifies that different modules interact properly, such as admission data flowing into analytics and evaluation modules.
- **System Testing:** Tests the entire system as a whole to ensure all components work together seamlessly.
- **User Acceptance Testing (UAT):** End users (students, staff, admin) test the system to confirm usability and correctness.
- **Performance Testing:** Checks system behavior under multiple users and heavy operations like test submissions and timetable generation.
- **Security Testing:** Ensures login authentication, role-based access, and data protection mechanisms are functioning properly.



### 8.1.1 Black Box Test Cases

| S.No | Use Case               | Input                | Expected Output          | Actual Output           | Status |
|------|------------------------|----------------------|--------------------------|-------------------------|--------|
| 1    | Login                  | Valid credentials    | User authenticated       | User logged in          | Pass   |
| 2    | Login                  | Invalid credentials  | Access denied            | Access denied           | Pass   |
| 3    | Admission Registration | Valid student data   | Registration success     | Registered successfully | Pass   |
| 4    | Document Verification  | Valid documents      | Verified successfully    | Verified                | Pass   |
| 5    | Seat Allocation        | Cutoff + preferences | Seat allocated correctly | Allocated               | Pass   |
| 6    | Timetable Generation   | Staff & subjects     | Conflict-free timetable  | Generated correctly     | Pass   |
| 7    | Test Submission        | Valid answers        | Marks generated          | Marks calculated        | Pass   |
| 8    | AI Evaluation          | Answer script PDF    | Marks assigned using OCR | Evaluated correctly     | Pass   |
| 9    | Performance Analysis   | Student data         | Charts displayed         | Charts generated        | Pass   |
| 10   | Coin System            | Test performance     | Coins awarded            | Coins updated           | Pass   |

**Table 7.1 Black Box Test Cases**

## 8.2 WHITE BOX TESTING

White box testing focuses on testing the internal logic, algorithms, and code structure of the system. It ensures that all internal operations such as data processing, condition checking, and algorithm execution are working correctly. This type of testing is particularly important for validating complex modules like timetable generation, AI evaluation, and data analysis.

### Types of White Box Testing Used:

- **Unit Testing:** Individual functions such as cutoff calculation, keyword matching, and data processing are tested independently.
- **Code Path Testing:** Ensures all logical paths and conditions in the code are executed and validated.
- **Loop Testing:** Verifies loops used in algorithms such as seat allocation, timetable generation, and evaluation processes.
- **Algorithm Testing:** Validates correctness of algorithms used in admission processing, scheduling, and evaluation modules.
- **Data Flow Testing:** Ensures proper flow of data between components such as frontend, backend, and database.

### 8.2.1 White Box Test Cases

| S.No | Module          | Test Scenario            | Expected Result            | Status |
|------|-----------------|--------------------------|----------------------------|--------|
| 1    | Admission       | Cutoff calculation logic | Correct cutoff generated   | Pass   |
| 2    | Admission       | Seat allocation loop     | Proper seat assignment     | Pass   |
| 3    | Timetable       | Constraint validation    | No conflicts in schedule   | Pass   |
| 4    | Timetable       | Staff availability check | No overlapping assignments | Pass   |
| 5    | Test Evaluation | MCQ validation logic     | Correct answer matching    | Pass   |
| 6    | AI Evaluation   | Keyword matching         | Accurate scoring           | Pass   |
| 7    | Data Processing | Data validation          | Clean and valid data       | Pass   |
| 8    | Analytics       | Performance calculation  | Correct metrics generated  | Pass   |
| 9    | Authentication  | Role validation logic    | Proper access control      | Pass   |
| 10   | Database        | Data storage & retrieval | Data stored correctly      | Pass   |

**Table 7.2 White Box Test Cases**

## CHAPTER 9

### CONCLUSION & FUTURE WORK

#### 9.1 CONCLUSION

The Web-Based Smart ERP Solution for Academic Institutions successfully provides an integrated platform for managing academic and administrative activities efficiently. By combining multiple modules such as admission management, administrative control, HOD management, connection platform, AI-based evaluation, and analytical module, the system simplifies complex institutional processes and reduces manual workload. The automation of tasks such as student registration, seat allocation, timetable generation, and evaluation improves accuracy, transparency, and operational efficiency. The system enhances communication between students, staff, and administrators through a centralized platform, ensuring smooth information flow. The AI-based evaluation module reduces the effort required for manual grading, while the analytical module provides valuable insights into student performance, enabling data-driven decision-making. The use of modern technologies such as React, Spring Boot, and MySQL ensures scalability, reliability, and a user-friendly experience. Overall, this project demonstrates how automation, artificial intelligence, and data analytics can transform traditional academic management systems into efficient, intelligent, and scalable solutions, improving both institutional productivity and educational outcomes.

## 9.2 FUTURE WORK

The proposed Smart ERP System can be further enhanced by incorporating advanced features to improve functionality, scalability, and user experience. A dedicated mobile application can be developed to provide seamless access for students and staff, enabling them to manage academic activities anytime and anywhere. The system can be upgraded with advanced AI and machine learning models to improve the accuracy of evaluation and enable predictive analytics for identifying student performance trends and risks. Deploying the system on cloud platforms can enhance scalability, availability, and performance, allowing it to handle large numbers of users efficiently. Real-time notification mechanisms such as push notifications, emails, and SMS alerts can be integrated to keep users informed about important updates like admissions, exams, and results. The system can also be extended to support multiple institutions within a single platform, making it suitable for large-scale academic management. Additionally, advanced analytics dashboards with predictive insights, behavioral analysis, and performance forecasting can be implemented to support better decision-making. Integration with external systems such as Learning Management Systems (LMS), digital libraries, and third-party APIs can further enhance the system's capabilities. Future improvements may also include enhanced security features,

biometric authentication, blockchain-based record management for secure data storage, and AI-driven personalized learning recommendations, making the system more intelligent, secure, and adaptable to evolving educational needs.